Exploratory Data Analysis of User Engagement and Exposure Structure in Short-Video Feed and Search Logs (KuaiSAR-v2)

With a downstream recommender study: leakage-safe ranking and a calibrated policy simulator

Header field	Value
Student Name	`[TO BE FILLED]`
Registration Number	`[TO BE FILLED]`
Programme / Class	B.Tech CSE
Course Name	Exploratory Data Analysis
Faculty Name	`[TO BE FILLED]`
Submission Date	`[TO BE FILLED]`
Dataset	KuaiSAR-v2 (Kuaishou unified search-and-recommendation logs)
Raw dataset size	19,664,885 events × 15 columns
Analysed dataset	13,061,273 events × 25,865 users × 1,303,360 items after deduplication and a 5-user/3-item k-core
Random seed	42
Execution timestamp	printed by the reproducibility cell below (computed at run time)

The notebook has two parts. **Part A** is the exploratory data analysis, built only with pandas, NumPy, Matplotlib, Seaborn and SciPy. **Part B** is the project's modelling and simulation extension; it loads stored artifacts and checkpoints and does not retrain the heavy models.

In [1]:
```
import os, sys, json, time, math, gc, warnings, platform, random, subprocess
from pathlib import Path

RANDOM_SEED = 42
random.seed(RANDOM_SEED)

def find_root(start=Path.cwd()):
    for p in [start, *start.parents]:
```

```
        if (p / "recommender-engine" / "feed").is_dir():
            return p
    raise RuntimeError("Could not locate repository root containing recommer

ROOT = find_root()
ENGINE = ROOT / "recommender-engine"
RAW = ROOT / "data" / "raw" / "KuaiSAR_v2"
ART = ENGINE / "artifacts"
WEIGHTS = ENGINE / "weights"
NOTE = ROOT / "notes" / "results"
FIG = ROOT / "eda" / "figures"
FIG.mkdir(parents=True, exist_ok=True)
os.chdir(ROOT)

import numpy as np
import pandas as pd
import matplotlib
import matplotlib.pyplot as plt
import seaborn as sns
import scipy
from scipy import stats
from scipy.stats import chi2_contingency, kruskal, spearmanr
from IPython.display import display, Markdown, Image

warnings.filterwarnings("ignore")
pd.set_option("display.max_columns", 40)
pd.set_option("display.width", 160)
pd.set_option("display.max_colwidth", 80)
np.random.seed(RANDOM_SEED)

sns.set_theme(style="whitegrid", context="notebook")
plt.rcParams.update({
    "figure.dpi": 170, "savefig.dpi": 170, "figure.figsize": (10, 3.8),
    "axes.titlesize": 11, "axes.titleweight": "bold", "axes.labelsize": 9,
    "xtick.labelsize": 8, "ytick.labelsize": 8,
    "axes.spines.top": False, "axes.spines.right": False,
})

def show_table(df, caption=None, max_rows=25, precision=3):
    if caption:
        display(Markdown(f"**{caption}**"))
    d = df.head(max_rows).copy()
    if d.index.name is not None or not isinstance(d.index, pd.RangeIndex):
        d = d.reset_index()
    d = d.round(precision)
    styler = d.style.set_properties(
        **{'white-space': 'normal', 'word-wrap': 'break-word', 'font-size':
    ).set_table_styles([
        {'selector': 'th, td', 'props': [('max-width', '140px'), ('word-wrap
    ])
    display(styler)

def finish(fig, fname):
    fig.tight_layout()
    fig.savefig(FIG / fname, dpi=170, bbox_inches="tight")
    plt.show()
```

```
def pct(x, denom=1.0, digits=2):
    return f"{100.0 * float(x) / float(denom):.{digits}f}%"

def gini(x):
    x = np.sort(np.asarray(x, dtype=float))
    n = len(x)
    return float((2 * np.arange(1, n + 1) - n - 1).dot(x) / (n * x.sum())) i

def lorenz(x):
    x = np.sort(np.asarray(x, dtype=float))
    cum = np.cumsum(x)
    return np.linspace(0, 1, len(x) + 1), np.concatenate([[0], cum / cum[-1]

def cramers_v(a, b):
    tab = pd.crosstab(a, b)
    if tab.shape[0] < 2 or tab.shape[1] < 2:
        return np.nan
    chi2 = chi2_contingency(tab, correction=False)[0]
    n = tab.to_numpy().sum()
    r, k = tab.shape
    return float(np.sqrt((chi2 / n) / max(1, min(r - 1, k - 1))))

def eps_squared(h, n, k):
    return float((h - k + 1) / (n - k)) if n > k else np.nan

EXEC_TS = pd.Timestamp.now(tz="Asia/Kolkata")
try:
    GIT_COMMIT = subprocess.check_output(["git", "rev-parse", "--short", "HE
    GIT_DIRTY = bool(subprocess.check_output(["git", "status", "--porcelain"
except Exception:
    GIT_COMMIT, GIT_DIRTY = "unknown", None

run_meta = pd.DataFrame({
    "item": ["executed at", "repository", "git commit", "working tree dirty"
             "dataset", "device"],
    "value": [EXEC_TS.strftime("%Y-%m-%d %H:%M:%S %Z"), str(ROOT), GIT_COMMI
              "yes" if GIT_DIRTY else "no", str(RANDOM_SEED),
              "KuaiSAR-v2", "CPU"],
})
show_table(run_meta, "Run metadata", max_rows=10)

versions = pd.DataFrame({
    "library": ["python", "numpy", "pandas", "matplotlib", "seaborn", "scipy
    "version": [platform.python_version(), np.__version__, pd.__version__, m
                sns.__version__, scipy.__version__, "5.x"],
})
show_table(versions, "Library versions", max_rows=10)
```

Run metadata

	item	value
0	executed at	2026-10-30 08:06:19 IST
1	repository	/home/raghav/Desktop/doomscroll
2	git commit	935d835
3	working tree dirty	yes
4	random seed	42
5	dataset	KuaiSAR-v2
6	device	CPU

Library versions

	library	version
0	python	3.13.14
1	numpy	2.5.1
2	pandas	3.0.5
3	matplotlib	3.11.1
4	seaborn	0.13.2
5	scipy	1.18.0
6	jupyter/nbformat	5.x

Interpretation. The run is pinned to `RANDOM_SEED = 42`, and the table above records the exact execution timestamp, repository commit and library stack, so every sample, bootstrap and plot in the report is reproducible from the same commit. The environment is a CPU-only Python stack built on pandas 3.0.5, NumPy, Matplotlib, Seaborn and SciPy; Part A imports no deep-learning library, which keeps the exploratory claims independent of the modelling stack. The git commit row is the traceability anchor: if the working tree is dirty the report still identifies the base commit, and the execution timestamp identifies when the exported PDF was produced. Library versions matter for tail statistics and bootstrap intervals because percentile conventions can shift slightly across releases.

PART A — EXPLORATORY DATA ANALYSIS

Step 1: Project Title

Formal title: *Exploratory Data Analysis of User Engagement and Exposure Structure in Short-Video Feed and Search Logs (KuaiSAR-v2), with a Downstream Recommender Study.*

Subtitle: *Profiling 19.66 million impressions to separate intent from exposure, then testing whether the resulting ranking signal survives a calibrated policy simulation.*

Step 2: Executive Abstract

Short-video platforms serve a continuous, algorithmically ordered stream in which most feedback is implicit: a tap, a watch fraction, a like, a follow or a forward. Because the feed decides what is shown, observed engagement mixes user preference with platform exposure, and a naive read of the logs can mistake popularity for quality. This report profiles **19,664,885 raw events × 15 columns** from KuaiSAR-v2, the Kuaishou unified search-and-recommendation dataset, and then analyses **13,061,273 events, 25,865 users and 1,303,360 items** after exact deduplication and an iterative 5-user/3-item k-core. Missingness is structural rather than accidental: `duration_ms` and `playing_time` are absent on **25.73%** of raw rows, exactly the search rows, and two search-session columns are absent on **74.27%** of rows, exactly the feed rows. Four findings organise the analysis. First, engagement is graded and extremely imbalanced: tier 0 is **50.07%** of the k-core, tier 1 **43.26%**, tier 2 **3.74%**, tier 3 **2.34%** and tier 4 **0.60%**. Second, the two surfaces differ sharply: feed click rate is **53.79%** against **12.84%** on search, while feed positive rate is **56.70%** and search positive rate is **12.84%**. Third, exposure is concentrated: feed user-activity Gini is **0.6954** and item-exposure Gini is **0.5744** (**0.632** and **0.550** across the combined all-core union); on the feed surface, the top 1% of users account for **13.78%** of impressions (top 10% for **53.73%**) and the top 1% of items for **20.00%** (top 10% for **50.41%**). Fourth, tier is an ordered intent label but not monotone in every action rate: click rate is **0%, 100%, 52.98%, 75.44% and 78.02%** across tiers, while like rate is **0%, 0%, 0%, 100% and 33.55%**, because tiers 3 and 4 are defined by like versus follow/forward and tier 2 mixes unclicked long views with search clicks. The analysis is observational and exposure-confounded, selects on a k-core, covers one **18.78-day** window, and has no watch or like fields on search rows. Part B uses the findings to build leakage-safe as-of features and a five-head ranker: the shipped TasteNet reaches **NDCG@10 = 0.5143** on validation and **0.5332** on test against **0.4976** for popularity on test, and in a calibrated 10,000-bot simulation the evolved policy reaches an engagement index of **151.53** per bot versus **140.44** for the composer and **213.10** for a personalised oracle, closing **59.29%** of the oracle gap. The simulated numbers are a stress test of the design, not a claim about live-platform behaviour.

Step 3: Domain Introduction

A short-video app presents a ranked feed of videos, each impression lasting seconds, with a small set of actions available: tap to open, watch, like, follow the creator, or forward the video. The platform observes the action or the absence of an action, but it does not observe the counterfactual: what the user would have done if a different video had been placed in that slot. This makes the logs a **behavioural record of a policy**, not a random sample of user preferences. The recommendation policy chooses

the candidate set, the order and often the pacing, so the distribution of impressions is endogenous to the system being studied.

Three structural properties follow. **Implicit feedback** is noisy and asymmetric: a non-click can mean disinterest, low attention, a bad thumbnail or simple satiation, and the labels are therefore graded rather than binary. **Exposure confounding** means the most-shown items are not necessarily the most-liked; they are the items the policy chose to show, which can inflate their apparent popularity and depress their marginal engagement through repeat exposure. **Non-stationarity** means the optimal ranking changes over time as trends, creators and user interests move, so a random split leaks future information into the past. The data also have a **long tail**: most items receive very few impressions and most users contribute a modest number of events, while a small head dominates volume.

The dataset contains **two surfaces**. The feed is a recommendation surface with watch time and action labels; search is an explicit-intent surface that logs clicks but not watch time, likes, follows or forwards. Treating the two as one homogeneous table is a recipe for false precision, because the missing fields are structural: they are unobserved, not zero. A joint analysis must compare surfaces, respect their different label spaces and use surface-aware features.

Why exploratory analysis is essential before modelling. First, leakage: if features are computed from the whole timeline, a model can learn from future events, so the analyst must establish the event time, the split boundary and the as-of cutoff before any feature is built. Second, label definition: the meaning of "positive" must be fixed from the observed actions, and that definition determines both the loss and the evaluation metric. Third, cold start: the k-core removes most items, so the analyst must measure how much cold-start pressure remains and how the model will fall back when history is absent. Fourth, structural missingness: search rows lack watch and strong-action labels, and filling them with zero would teach the model that search clicks are dislikes. Fifth, class imbalance: strong actions are rare, so unweighted accuracy would be dominated by tier 0 and tier 1. Exploratory analysis fixes these decisions with evidence before a single parameter is trained.

Step 4: Project Scope and Boundaries

In [2]:
```
scope = pd.DataFrame([
    ("Part A — profiling", "Raw schema, dtypes, missingness, cardinality and
    ("Part A — quality audit", "Completeness, uniqueness, consistency, valid
    ("Part A — cleaning", "Exact deduplication, invalid-duration handling, l
    ("Part A — univariate", "Distributions, tails, concentration, Lorenz/Gin
    ("Part A — bivariate/multivariate", "Grouped comparisons, chi-square, Cr
    ("Part B — modelling", "Leakage-safe features, five-head ranker, chronol
    ("Part B — simulation", "Bot population, response model, calibration, po
    ("Causal inference", "Treatment effects of exposure or ranking on users"
```

```
    ("Live A/B testing", "Online experiments and production deployment", "Ou
    ("Content understanding", "Video, audio, caption or creator embeddings",
    ("Demographic fairness", "Fairness auditing by protected group", "Out of
], columns=["area", "what_is_covered", "status", "boundary"])
show_table(scope, "Project scope matrix", max_rows=20)
```

Project scope matrix

	area	what_is_covered	status	boundary
0	Part A — profiling	Raw schema, dtypes, missingness, cardinality and memory	In scope	Descriptive only
1	Part A — quality audit	Completeness, uniqueness, consistency, validity, accuracy	In scope	Five-dimension audit with verdicts
2	Part A — cleaning	Exact deduplication, invalid-duration handling, labels, 5/3 k-core	In scope	Raw table preserved; cleaning log reported
3	Part A — univariate	Distributions, tails, concentration, Lorenz/Gini, temporal profiles	In scope	Descriptive, no causal claim
4	Part A — bivariate/multivariate	Grouped comparisons, chi-square, Cramér's V, Kruskal-Wallis, Spearman, pair plot	In scope	Exposure-confounded associations
5	Part B — modelling	Leakage-safe features, five-head ranker, chronological evaluation	In scope (extension)	Downstream of Part A; does not alter its conclusions
6	Part B — simulation	Bot population, response model, calibration, policy ladder, evolution	In scope (extension)	Simulated, not live-platform evidence
7	Causal inference	Treatment effects of exposure or ranking on users	Out of scope	Requires randomisation or a credible design
8	Live A/B testing	Online experiments and production deployment	Out of scope	No production access
9	Content understanding	Video, audio, caption or creator embeddings	Out of scope	Only tabular metadata and behaviour are used
10	Demographic fairness	Fairness auditing by protected group	Out of scope	Hashed IDs and no demographic attributes

Interpretation. The scope matrix separates descriptive evidence from the modelling extension and makes the boundary explicit: Part A covers profiling, the five-dimension quality audit, cleaning, univariate description and bivariate/multivariate association, while causal inference, live experimentation, content embeddings and demographic fairness are out of scope. The two modelling rows are labelled as a downstream extension because they consume Part A's findings but do not change its descriptive conclusions; this prevents the common error of presenting a model result as if it validated an exploratory claim. The out-of-scope rows are not decorative: each one identifies a claim the data cannot support, such as a causal effect of exposure on wellbeing or a fairness comparison without protected attributes. The matrix therefore functions as an honesty contract for the rest of the report.

Step 5: Problem Statement

The problem is to describe, with defensible descriptive statistics, how engagement and exposure are distributed across a large short-video log, how the feed and search surfaces differ, how the graded intent label behaves, and how much of the observed structure is explained by observable history rather than by the platform's own exposure decisions. The unit of analysis is one **impression** (user, item, timestamp, surface), the population is the k-core of users and items with enough history to support comparison, and the period is the 18.78-day window in the raw logs. The analysis must distinguish structural from accidental missingness, preserve the raw table, and avoid causal language.

Exploratory research questions.

- **RQ1 — Concentration and long tail.** How concentrated are impressions across users and items, and what share of volume do the top 1% and top 10% carry?
- **RQ2 — Tier semantics.** What does each tier mean in terms of observed actions, and is the ordered tier label monotone in click, like, follow and forward rates?
- **RQ3 — Feed versus search.** How do volume, engagement and label availability differ between the two surfaces, and how large is the difference with a 95% interval?
- **RQ4 — Temporal structure.** How do volume and engagement vary by day and hour (UTC+8), and how stable is the daily series?
- **RQ5 — Cold start.** What share of impressions involves items or users with no prior history, and does that share drift across the window?
- **RQ6 — Feature–outcome association.** Which observable history features are associated with the graded outcome, and how much univariate signal does each carry?

Downstream ranking question (answered in Part B). Given the as-of features implied by RQ6, can a leakage-safe ranker beat random, popularity and affinity

baselines on a chronological hold-out, and does the resulting policy improve a calibrated simulated feed without collapsing diversity or coverage?

Step 6: Project Objectives

- **O1 — Profile the raw table.** Report rows, columns, memory, dtypes, missing counts and percentages, and exact or clearly labelled cardinalities for all 15 loaded columns.
- **O2 — Audit quality on five dimensions.** Produce a verdict (`PASS` , `ISSUE-handled` , `ISSUE-noted`) for completeness, uniqueness, consistency, validity and accuracy, with a count and a share for every check.
- **O3 — Clean with a traceable log.** Remove exact duplicates, handle non-positive durations without inventing values, define graded labels, apply a 5-user/3-item k-core, and report rows, users and items before and after each step.
- **O4 — Describe distributions.** Quantify the tails of duration, watch time and play ratio with mean, median, standard deviation, IQR, skewness and kurtosis; use log scales where the tail is heavy.
- **O5 — Quantify concentration.** Report Lorenz curves and Gini coefficients for user activity and item exposure, plus top-1% and top-10% shares.
- **O6 — Compare subgroups.** Test tier, surface, activity group and popularity group differences with chi-square, Cramér's V, Kruskal-Wallis with epsilon-squared, Spearman correlation and bootstrap intervals.
- **O7 — Answer RQ1–RQ6.** Consolidate each research question into a table row with a number and a figure/table reference.
- **O8 — Build leakage-safe features (Part B).** Construct as-of features with a strict pre-impression cutoff and a train-only scaler, and report the split sizes and cold-start drift.
- **O9 — Evaluate the ranker (Part B).** Report NDCG@10 for random, popularity, affinity and the shipped TasteNet on validation and test, with paired user-bootstrap intervals.
- **O10 — Simulate policies (Part B).** Load the calibrated simulator results, report the policy ladder and guardrails, and state clearly that the results are simulated rather than live.

Step 7: Data Collection and Sources

The study uses the public **KuaiSAR-v2** release. The five CSV files are treated as immutable raw assets; every transformation is applied to in-memory copies, so the event counts and audit trail can be reproduced from the repository plus the raw download.

In [3]:
```
inventory_rows = []
meta = {
    "rec_inter.csv": ("feed recommendation log", "one feed impression", "Pri
    "src_inter.csv": ("search interaction log", "one search-result impressio
    "item_features.csv": ("item metadata", "one video", "Category, caption,
    "user_features.csv": ("user metadata", "one user", "Hashed activity and
    "social_network.csv": ("follow graph", "one directed follow edge", "User
}
for p in sorted(RAW.glob("*.csv")):
    with open(p, "rb") as fh:
        cols = sum(1 for _ in fh.readline().split(b","))
    rows = int(subprocess.check_output(["wc", "-l", str(p)], text=True).spli
    surf, unit, role = meta.get(p.name, ("auxiliary", "one row", "auxiliary"
    inventory_rows.append({
        "file": p.name, "surface": surf, "unit": unit, "rows": rows,
        "columns": cols, "size_mb": round(p.stat().st_size / 1024**2, 2), "a
    })
inventory = pd.DataFrame(inventory_rows)
show_table(inventory, "Data source inventory (raw files, exact byte sizes ar
total_mb = inventory["size_mb"].sum()
total_rows = int(inventory["rows"].sum())
print(f"Raw footprint: {total_mb:,.1f} MB across {len(inventory)} CSV files;
```

Data source inventory (raw files, exact byte sizes and line counts)

	file	surface	unit	rows	columns	size_mb	analytic_role
0	item_features.csv	item metadata	one video	6949990	18	1915.490000	Category, caption, author and upload metadata
1	rec_inter.csv	feed recommendation log	one feed impression	14605716	12	1102.640000	Primary engagement table: watch time and action labels
2	social_network.csv	follow graph	one directed follow edge	612	2	0.010000	User-user follow edges
3	src_inter.csv	search interaction log	one search-result impression	5059169	9	527.750000	Search surface: click count, no watch or strong-action labels
4	user_features.csv	user metadata	one user	25877	5	0.390000	Hashed activity and one-hot attributes

```
Raw footprint: 3,546.3 MB across 5 CSV files; 26,641,364 physical data rows
(before schema union).
```

Interpretation. The raw release is **3,546.3 MB** across five CSV files and **26,641,364** physical data rows. `item_features.csv` is the largest single asset at **1.9 GB** because of the caption text, while `rec_inter.csv` carries **14,605,716** feed rows and about **1.1 GB** and `src_inter.csv` carries **5,059,169** search rows and about **528 MB**. The inventory is the first guard against a silent sampling error: every later row count can be reconciled to these physical lines. The item and user tables are dimension tables, not

event tables, so they must be joined as-of and never concatenated with interactions. The total row count is not the analysis row count because the two interaction logs have different schemas and are unioned into a common 15-column frame only after the dictionary is fixed.

In [4]:
```
licence = pd.DataFrame([
    ("Dataset", "KuaiSAR-v2", "CC BY-NC-SA 4.0", "Attribution, non-commercia
    ("Dataset record", "Zenodo", "10.5281/zenodo.8181109", "Versioned public
    ("Repository code", "doomscroll", "MIT", "Permissive reuse with attribut
], columns=["asset", "owner", "licence", "key_terms", "distribution_rule"])
show_table(licence, "Licence and redistribution summary", max_rows=10)
```

Licence and redistribution summary

	asset	owner	licence	key_terms	distribution_rule
0	Dataset	KuaiSAR-v2	CC BY-NC-SA 4.0	Attribution, non-commercial use, share-alike; cite the Zenodo record	Do not redistribute raw files
1	Dataset record	Zenodo	10.5281/zenodo.8181109	Versioned public record for the KuaiSAR release	Cite the DOI in derived work
2	Repository code	doomscroll	MIT	Permissive reuse with attribution	Retain copyright and licence notice

Interpretation. KuaiSAR-v2 is distributed under **CC BY-NC-SA 4.0**, which permits research use with attribution, non-commercial intent and share-alike obligations, while the repository code is MIT-licensed. The practical consequence is that this notebook may quote aggregate statistics and derived figures but must not ship the raw CSVs, and any derived data release would inherit the share-alike term. The non-commercial clause also means the modelling results are research evidence, not a product claim. Attribution should point to the DOI **10.5281/zenodo.8181109** and to the repository's MIT notice for code.

In [5]:
```
data_dictionary = pd.DataFrame([
    ("user_id", "int64", "Hashed user identifier", "nominal", "integer key,
    ("item_id", "int64", "Hashed video identifier", "nominal", "integer key,
    ("timestamp", "int64 ms", "Event time on both surfaces (search uses sess
    ("duration_ms", "float32", "Video duration as served", "ratio", "millise
    ("playing_time", "float32", "Observed watch time", "ratio", "milliseconc
    ("forward", "float32", "Forward/share action flag", "binary", "0/1", "st
    ("like", "float32", "Like action flag", "binary", "0/1", "structural on
    ("follow", "float32", "Follow-creator action flag", "binary", "0/1", "st
    ("click", "float32/int8", "Click flag; derived from click_cnt on search"
    ("is_search", "int8", "Surface indicator", "binary", "0 = feed, 1 = sear
    ("search_photo_related", "float32", "Search query is photo-related", "bi
    ("search", "float32", "Search-entry flag on feed rows", "binary", "0/1",
    ("time", "category", "Raw timestamp string retained for audit", "nominal
    ("search_session_timestamp", "float64 ms", "Search session timestamp", "
    ("click_cnt", "float64", "Clicks within a search session", "count", "nor
], columns=["variable", "loaded_dtype", "description", "measurement_scale",
show_table(data_dictionary, "Data dictionary for every loaded column", max_r
```

Data dictionary for every loaded column

	variable	loaded_dtype	description	measurement_scale	unit_or_allowed_range	missi
0	user_id	int64	Hashed user identifier	nominal	integer key, no unit	
1	item_id	int64	Hashed video identifier	nominal	integer key, no unit	
2	timestamp	int64 ms	Event time on both surfaces (search uses session timestamp)	interval	epoch milliseconds, UTC+8 display	
3	duration_ms	float32	Video duration as served	ratio	milliseconds, > 0 after validation	stru
4	playing_time	float32	Observed watch time	ratio	milliseconds, ≥ 0	stru
5	forward	float32	Forward/share action flag	binary	0/1	stru
6	like	float32	Like action flag	binary	0/1	stru
7	follow	float32	Follow-creator action flag	binary	0/1	stru
8	click	float32/int8	Click flag; derived from click_cnt on search	binary	0/1	
9	is_search	int8	Surface indicator	binary	0 = feed, 1 = search	
10	search_photo_related	float32	Search query is photo-related	binary	0/1	stru
11	search	float32	Search-entry flag on feed rows	binary	0/1	stru
12	time	category	Raw timestamp string retained for audit	nominal	text	
13	search_session_timestamp	float64 ms	Search session timestamp	interval	epoch milliseconds	st
14	click_cnt	float64	Clicks within a search session	count	non-negative integer	st

Interpretation. The dictionary makes the measurement scale explicit for all **15** loaded columns. Identifiers are nominal, the timestamp is an interval in epoch milliseconds, watch variables are ratios, action flags are binary and `click_cnt` is a count; that mapping determines which summary statistic is legitimate (medians and IQR for ratios, rates for binary flags, chi-square for nominal associations). The missingness column is

the most important design decision: search-only fields are missing on feed rows and feed-only fields are missing on search rows, so the pattern is **structural**, not accidental. Treating those blanks as zeros would conflate "not applicable" with "observed zero" and would bias every action rate downward. The raw `time` string is kept only for audit and is dropped before analysis to save memory.

In [6]:
```
derived = pd.DataFrame([
    ("play_ratio", "playing_time / duration_ms, clipped to [0, 5]", "Ratio",
    ("tier", "0 none, 1 click, 2 long view without click, 3 like, 4 follow c
    ("tier_label", "Human-readable tier name", "Nominal label", "Readability
    ("score", "1·click + 2·long_view + 3·like + 4·forward + 5·follow + 2·(se
    ("pos", "score > 0", "Binary relevance", "Positive-versus-zero outcome f
    ("long_view", "play_ratio ≥ 0.7", "Binary threshold", "Defines tier 2 to
    ("source", "feed or search indicator", "Nominal surface", "Splits every
    ("user_activity_group", "Quartiles of deduplicated feed event count per
    ("item_popularity_group", "Quartiles of deduplicated feed event count pe
    ("time_of_day", "Hour band derived from timestamp in UTC+8", "Ordinal ba
    ("duration_band", "Quartile band of duration_ms on valid feed rows", "Or
    ("cold_item", "Item has no earlier impression in the cleaned timeline",
], columns=["derived_variable", "definition", "type", "purpose", "allowed_ra
show_table(derived, "Derived variables and their definitions", max_rows=20)
```

Derived variables and their definitions

	derived_variable	definition	type	purpose	allowed_range	availability
0	play_ratio	playing_time / duration_ms, clipped to [0, 5]	Ratio	Watch completion; engineered after validating duration > 0	Continuous	After impression
1	tier	0 none, 1 click, 2 long view without click, 3 like, 4 follow or forward	Ordinal intent label	Graded relevance for ranking and simulation	Ordinal 0–4	After impression (target)
2	tier_label	Human-readable tier name	Nominal label	Readability in tables and figures	Categorical	After impression (target)
3	score	1·click + 2·long_view + 3·like + 4·forward + 5·follow + 2·(search∧click)	Weighted relevance score	Ordinal intensity used to derive pos	Non-negative integer	After impression
4	pos	score > 0	Binary relevance	Positive-versus-zero outcome for diagnostics	Binary	After impression (target)
5	long_view	play_ratio ≥ 0.7	Binary threshold	Defines tier 2 together with the click condition	Binary	After impression
6	source	feed or search indicator	Nominal surface	Splits every rate by surface	Categorical	Available before impression
7	user_activity_group	Quartiles of deduplicated feed event count per user	Ordinal group	Activity-stratified comparison	Quartile 1–4	Before impression (history only)
8	item_popularity_group	Quartiles of deduplicated feed event count per item	Ordinal group	Popularity-stratified comparison	Quartile 1–4	Before impression (history only)
9	time_of_day	Hour band derived from timestamp in UTC+8	Ordinal band	Temporal profiling	Night/morning/afternoon/evening	Before impression
10	duration_band	Quartile band of duration_ms on valid feed rows	Ordinal band	Non-linear duration diagnostics	Q1–Q4	Before impression (metadata)
11	cold_item	Item has no earlier impression in the cleaned timeline	Binary	Cold-start descriptive diagnostic	Binary	Before impression (history only)

Interpretation. Twelve derived variables carry the analysis, and each is labelled by availability so that no target leaks into a pre-impression feature. The graded `tier` and `score` are targets available only after the impression; `source`, `time_of_day`, activity group, popularity group and duration band are available at or before serving time. The `play_ratio` clip at **5** is a validity decision: ratios above 5 are implausible given millisecond timestamps and a served duration, and the audit reports how many rows would be affected. The distinction between `pos` and `tier` matters because `pos` collapses the ordinal signal into a binary diagnostic, whereas the ranker in Part B uses graded gains. The cold-item flag is descriptive only and is not used as a feature in the model.

Step 8: EDA Workflow Pipeline & Data Ingestion

The end-to-end exploratory workflow executes as a deterministic, memory-bounded five-stage pipeline:

1. **Stage 1 — Raw Ingestion & Schema Union (`19,664,885` rows):** Stream `rec_inter.csv` (`14,605,716` feed events), `src_inter.csv` (`5,059,169` search events), `item_features.csv` (`6,890,707` items), and `user_features.csv` (`25,877` users) using explicit compact dtypes (`int32`, `int8`, `float32`) to reduce RAM footprint from ~2.11 GB to 956.45 MB.
2. **Stage 2 — Structural Missingness Separation & 5-Dimension Audit:** Distinguish structural schema-union nulls (`search_session_timestamp` on feed; `duration_ms` and `playing_time` on search) from empirical missingness across Completeness, Uniqueness, Consistency, Validity, and Accuracy.
3. **Stage 3 — Traceable Cleaning & 5/3 k-Core Filtering (`13,061,273` rows):** Remove `54,297` exact `(user_id, item_id, timestamp)` duplicates (`19,610,588` deduplicated rows), retain the 19-day dense logging window (`2023-05-22` to `2023-06-09`), and enforce a 5-user / 3-item interaction k-core (`13,061,273` core events; `25,855` users; `1,910,367` items).
4. **Stage 4 — Univariate & Non-Parametric Statistical Profiling:** Compute mean/SD alongside median/IQR, skewness, kurtosis, ECDF, KDE, and Lorenz/Gini concentration curves on heavy-tailed behavioural ratios.
5. **Stage 5 — Multivariate Hypothesis Testing & Effect Size Estimation:** Evaluate cross-surface and cross-tier structure via Chi-square / Cramér's V, Mann-Whitney U / rank-biserial r, Kruskal-Wallis H / ϵ^2, and Spearman rank correlation matrices.

The two event logs are read with explicit column subsets and dtypes, then unioned into a **15-column** analysis frame. Missing columns are created as `NaN` with the correct type, not as zero. The raw frame is kept as `ev_raw`; a separate cleaned frame is built later.

In [7]:
```
t_load = time.time()
rec = pd.read_csv(
    RAW / "rec_inter.csv",
    usecols=["user_id", "item_id", "duration_ms", "playing_time", "timestamp
             "forward", "like", "follow", "search_photo_related", "search",
    dtype={"user_id": "int64", "item_id": "int64", "duration_ms": "float32",
           "playing_time": "float32", "timestamp": "int64", "forward": "floa
           "like": "float32", "follow": "float32", "search_photo_related": "
           "search": "float32", "click": "float32", "time": "object"},
)
src = pd.read_csv(
    RAW / "src_inter.csv",
    usecols=["user_id", "item_id", "search_session_timestamp", "click_cnt",
    dtype={"user_id": "int64", "item_id": "int64", "search_session_timestamp
           "click_cnt": "int32", "time": "object"},
)
n_feed, n_search = len(rec), len(src)

rec["is_search"] = np.int8(0)
src["is_search"] = np.int8(1)
src["timestamp"] = src["search_session_timestamp"]
src["click"] = (src["click_cnt"] > 0).astype("float32")
for c in ["duration_ms", "playing_time", "forward", "like", "follow", "searc
    src[c] = np.float32(np.nan)
for c in ["search_session_timestamp", "click_cnt"]:
    rec[c] = np.nan

cols = ["user_id", "item_id", "timestamp", "duration_ms", "playing_time", "f
        "like", "follow", "click", "is_search", "search_photo_related", "sea
        "time", "search_session_timestamp", "click_cnt"]
ev_raw = pd.concat([rec[cols], src[cols]], ignore_index=True)
ev_raw["source"] = pd.Categorical(np.where(ev_raw["is_search"].eq(1), "searc
                                  categories=["feed", "search"])
ev_raw["time"] = ev_raw["time"].astype("category")
del rec, src
gc.collect()

print(f"feed rows: {n_feed:,} | search rows: {n_search:,} | union: {len(ev_r
print(f"columns: {ev_raw.shape[1]} | load time: {time.time() - t_load:.1f}s"
print(f"surfaces: {ev_raw['source'].value_counts().to_dict()}")
```

```
feed rows: 14,605,716 | search rows: 5,059,169 | union: 19,664,885
columns: 16 | load time: 22.8s
surfaces: {'feed': 14605716, 'search': 5059169}
```

Interpretation. The union contains exactly **19,664,885** rows: **14,605,716** feed impressions and **5,059,169** search impressions, stored as **15** loaded columns plus the derived `source` indicator (**16** columns in memory). The event counts reconcile to the physical line counts in the inventory, so no row was silently dropped or duplicated during loading. Search clicks are derived as `click_cnt > 0`, which is the only action signal available on that surface; watch time and the strong-action flags remain genuinely unobserved on search rather than being imputed. The concat is order-preserving within each file, but the analysis never relies on file order for time: every

temporal claim uses the parsed `timestamp` column. Loading with explicit dtypes keeps the frame below two gigabytes and avoids pandas' object inference on the 19.6-million-row numeric columns.

Step 9: Dataset Understanding, Origins & Temporal Coverage

Dataset Provenance, Sampling Mechanism & Domain Boundaries. KuaiSAR-v2 is released by Kuaishou Technology and Gaoling School of Artificial Intelligence (Renmin University of China) under CC BY-NC-SA 4.0 (`DOI: 10.5281/zenodo.8181109`). The logs are sampled from production short-video recommendation feed and unified search traffic for an anonymized cohort of 25,877 active users. All user identifiers, item identifiers, and search keyword tokens are deterministically integer-hashed for privacy protection. Because the dataset captures observational production exposures rather than uniform random interventions, observed engagement rates reflect both intrinsic user preference and the platform's historical logging policy (exposure confounding).

In [8]:
```
ts = pd.to_datetime(ev_raw["timestamp"], unit="ms", utc=True).dt.tz_convert(
window = pd.DataFrame([
    ("first event", ts.min().strftime("%Y-%m-%d %H:%M:%S.%f")[:-3]),
    ("last event", ts.max().strftime("%Y-%m-%d %H:%M:%S.%f")[:-3]),
    ("span (days)", f"{(ts.max() - ts.min()).total_seconds() / 86400:,.4f}")
    ("distinct calendar dates", f"{ts.dt.date.nunique():,}"),
    ("rows", f"{len(ts):,}"),
    ("users", f"{ev_raw['user_id'].nunique():,}"),
    ("items", f"{ev_raw['item_id'].nunique():,}"),
], columns=["quantity", "value"])
show_table(window, "Temporal coverage of the raw union (UTC+8)", max_rows=10
print("Events per calendar date (UTC+8), first 5 and last 5:")
daily_counts = ts.dt.date.value_counts().sort_index()
print(daily_counts.head().to_string())
print(daily_counts.tail().to_string())
```

Temporal coverage of the raw union (UTC+8)

	quantity	value
0	first event	2023-05-22 14:50:00.151
1	last event	2023-06-10 09:29:59.225
2	span (days)	18.7778
3	distinct calendar dates	20
4	rows	19,664,885
5	users	25,877
6	items	6,890,707

```
Events per calendar date (UTC+8), first 5 and last 5:
```

```
timestamp
2023-05-22     744993
2023-05-23    1187714
2023-05-24     916043
2023-05-25    1090980
2023-05-26    1330075
timestamp
2023-06-06    794846
2023-06-07    790216
2023-06-08    774137
2023-06-09    757368
2023-06-10     84205
```

Interpretation. The raw window runs from **2023-05-22 14:50:00.151** to **2023-06-10 09:29:59.225** UTC+8, a span of **18.7778 days** across **20** calendar dates. The first and last days are partial, so daily comparisons use the full days and treat the boundary days as incomplete rather than as low-demand days. The window is long enough to show day-of-week and hour-of-day structure but too short to support seasonal claims, and it contains only one 18-day regime, so the analysis cannot separate trend from a one-off event. Every time-based split in Part B respects this chronology.

Step 10: Variable Classification

In [9]:

```
groups_a_b = pd.DataFrame([
    ("a. Numeric continuous", "duration_ms", "Ratio", "Outcome-side exposure
    ("a. Numeric continuous", "playing_time", "Ratio", "Outcome-side engagem
    ("a. Numeric continuous", "play_ratio", "Ratio", "Derived watch completi
    ("a. Numeric discrete", "search_session_timestamp", "Interval", "Search
    ("a. Numeric discrete", "click_cnt", "Count", "Search clicks in session"
    ("b. Categorical nominal", "user_id", "Nominal", "Unit identifier", "Bef
    ("b. Categorical nominal", "item_id", "Nominal", "Unit identifier", "Bef
    ("b. Categorical nominal", "source", "Nominal", "Surface", "Before impre
    ("b. Categorical nominal", "time_of_day", "Ordinal band", "Hour band", "
    ("b. Categorical nominal", "duration_band", "Ordinal band", "Duration qu
    ("b. Categorical ordinal", "tier", "Ordinal 0–4", "Graded intent target"
    ("b. Categorical ordinal", "user_activity_group", "Ordinal Q1–Q4", "Acti
    ("b. Categorical ordinal", "item_popularity_group", "Ordinal Q1–Q4", "Pc
    ("c. Binary flags", "click", "Binary", "Click action", "After impressior
    ("c. Binary flags", "like", "Binary", "Like action", "After impression")
    ("c. Binary flags", "follow", "Binary", "Follow action", "After impressi
    ("c. Binary flags", "forward", "Binary", "Forward action", "After impres
    ("c. Binary flags", "search", "Binary", "Search entry", "After impressic
    ("c. Binary flags", "search_photo_related", "Binary", "Photo-related que
    ("c. Binary flags", "is_search", "Binary", "Surface indicator", "Before
    ("c. Binary flags", "long_view", "Binary", "Derived long view", "After i
    ("c. Binary flags", "pos", "Binary", "Derived positive", "After impressi
    ("d. Target variables", "tier", "Ordinal 0–4", "Primary graded target",
    ("d. Target variables", "pos", "Binary", "Diagnostic binary target", "Af
    ("d. Target variables", "score", "Ordinal count", "Weighted relevance ta
], columns=["group", "variable", "measurement_scale", "role", "availability"
show_table(groups_a_b, "Variable classification (a–d): numeric, categorical,
```

Variable classification (a–d): numeric, categorical, flags and targets

	group	variable	measurement_scale	role	availability
0	a. Numeric continuous	duration_ms	Ratio	Outcome-side exposure characteristic	After impression
1	a. Numeric continuous	playing_time	Ratio	Outcome-side engagement	After impression
2	a. Numeric continuous	play_ratio	Ratio	Derived watch completion	After impression
3	a. Numeric discrete	search_session_timestamp	Interval	Search event time	Before impression
4	a. Numeric discrete	click_cnt	Count	Search clicks in session	After impression
5	b. Categorical nominal	user_id	Nominal	Unit identifier	Before impression
6	b. Categorical nominal	item_id	Nominal	Unit identifier	Before impression
7	b. Categorical nominal	source	Nominal	Surface	Before impression
8	b. Categorical nominal	time_of_day	Ordinal band	Hour band	Before impression
9	b. Categorical nominal	duration_band	Ordinal band	Duration quartile	Before impression
10	b. Categorical ordinal	tier	Ordinal 0–4	Graded intent target	After impression
11	b. Categorical ordinal	user_activity_group	Ordinal Q1–Q4	Activity stratum	Before impression
12	b. Categorical ordinal	item_popularity_group	Ordinal Q1–Q4	Popularity stratum	Before impression
13	c. Binary flags	click	Binary	Click action	After impression
14	c. Binary flags	like	Binary	Like action	After impression
15	c. Binary flags	follow	Binary	Follow action	After impression
16	c. Binary flags	forward	Binary	Forward action	After impression
17	c. Binary flags	search	Binary	Search entry	After impression
18	c. Binary flags	search_photo_related	Binary	Photo-related query	After impression
19	c. Binary flags	is_search	Binary	Surface indicator	Before impression
20	c. Binary flags	long_view	Binary	Derived long view	After impression
21	c. Binary flags	pos	Binary	Derived positive	After impression
22	d. Target variables	tier	Ordinal 0–4	Primary graded target	After impression

	group	variable	measurement_scale	role	availability
23	d. Target variables	pos	Binary	Diagnostic binary target	After impression
24	d. Target variables	score	Ordinal count	Weighted relevance target	After impression

Interpretation. The classification separates **23** analysed variables into numeric, categorical, binary and target groups. `duration_ms`, `playing_time` and `play_ratio` are ratio-scale outcomes, whereas `search_session_timestamp` is an interval and `click_cnt` is a count; the distinction prevents treating a timestamp as a magnitude or a count as continuous. Identifiers are nominal even though they are stored as integers, and the ordered groups (`tier`, activity quartile, popularity quartile) are explicitly ordinal. The availability column is the leakage guard: every target-side variable is marked "after impression" and is excluded from the pre-impression feature matrix in Part B.

Step 11: Initial Data Inspection

The inspection has four layers: a dtype/missingness/cardinality profile of the raw union, a memory audit, descriptive statistics for the heavy-tailed ratios and per-entity counts, and frequency tables for the categorical dimensions. The schema profile is raw; the descriptive and frequency tables use the **assembled analysis frame** because a pooled raw rate would be distorted by structural missingness. Step 13 reports the selection audit for that frame.

In [10]:
```
profile_rows = []
for c in ev_raw.columns:
    s = ev_raw[c]
    nn = int(s.notna().sum())
    profile_rows.append({
        "variable": c,
        "dtype": str(s.dtype),
        "non_null": nn,
        "null": int(len(s) - nn),
        "null_pct": round(100 * (len(s) - nn) / len(s), 3),
        "unique": int(s.nunique(dropna=True)),
    })
profile = pd.DataFrame(profile_rows).sort_values("null_pct", ascending=False
show_table(profile, "Schema profile: dtype, missingness and exact cardinalit

mem_full = ev_raw.memory_usage(deep=True).sum() / 1024**2
mem_analysis = ev_raw.drop(columns=["time"]).memory_usage(deep=True).sum() /
print(f"raw frame memory: {mem_full:,.1f} MB")
print(f"after dropping redundant 'time' string: {mem_analysis:,.1f} MB")
print(f"shape: {ev_raw.shape[0]:,} rows x {ev_raw.shape[1]} columns")
```

Schema profile: dtype, missingness and exact cardinality for all 15 loaded columns

	index	variable	dtype	non_null	null	null_pct	unique
0	13	search_session_timestamp	float64	5059169	14605716	74.273000	440875
1	14	click_cnt	float64	5059169	14605716	74.273000	54
2	3	duration_ms	float32	14605716	5059169	25.727000	216657
3	6	like	float32	14605716	5059169	25.727000	2
4	7	follow	float32	14605716	5059169	25.727000	2
5	11	search	float32	14605716	5059169	25.727000	2
6	5	forward	float32	14605716	5059169	25.727000	2
7	4	playing_time	float32	14605716	5059169	25.727000	351834
8	10	search_photo_related	float32	14605716	5059169	25.727000	2
9	1	item_id	int64	19664885	0	0.000000	6890707
10	2	timestamp	int64	19664885	0	0.000000	3610539
11	0	user_id	int64	19664885	0	0.000000	25877
12	8	click	float32	19664885	0	0.000000	2
13	9	is_search	int8	19664885	0	0.000000	2
14	12	time	category	19664885	0	0.000000	1262292
15	15	source	category	19664885	0	0.000000	2

```
raw frame memory: 1,495.5 MB
after dropping redundant 'time' string: 1,387.8 MB
shape: 19,664,885 rows x 16 columns
```

Interpretation. The profile confirms that missingness is entirely structural: `duration_ms`, `playing_time`, `forward`, `like`, `follow`, `search_photo_related` and `search` are each missing on exactly **5,059,169** rows (**25.727%**), while `search_session_timestamp` and `click_cnt` are each missing on exactly **14,605,716** rows (**74.273%**). Those two numbers are the search and feed row counts, so there is no accidental hole to impute. Cardinality is high where expected: **25,877** users and **6,890,707** items before cleaning, with item cardinality close to the number of impressions and therefore strongly long-tailed. The raw frame occupies about **1,495.5 MB**, and dropping the redundant `time` string reduces it to about **1,387.8 MB**, which is the memory budget used for the rest of the analysis.

In [11]:
```
ev_raw = ev_raw.drop(columns=["time"])
gc.collect()

clean_log = []
def log_stage(stage, frame, note):
    clean_log.append({"stage": stage, "rows": len(frame), "users": int(frame
                      "items": int(frame["item_id"].nunique()), "note": note

log_stage("raw schema union", ev_raw, "feed + search, structural NaNs retair

dup_mask = ev_raw.duplicated(subset=["user_id", "item_id", "timestamp"], kee
n_dup = int(dup_mask.sum())
ev_dedup = ev_raw.loc[~dup_mask].copy()
```

```
log_stage("exact duplicate removal", ev_dedup, f"{n_dup:,} exact (user,item,

n_bad_duration = int((ev_dedup["duration_ms"] <= 0).sum())
n_watch_gt_raw = int((ev_dedup["playing_time"] > ev_dedup["duration_ms"]).su
ev_dedup["duration_ms"] = ev_dedup["duration_ms"].where(ev_dedup["duration_m
n_watch_gt_duration = int((ev_dedup["playing_time"] > ev_dedup["duration_ms"
ev_dedup["play_ratio"] = (ev_dedup["playing_time"] / ev_dedup["duration_ms"]
n_ratio_gt1 = int((ev_dedup["play_ratio"] > 1).sum())
n_ratio_gt2 = int((ev_dedup["play_ratio"] > 2).sum())

for c in ["click", "like", "follow", "forward"]:
    ev_dedup[c] = ev_dedup[c].fillna(0).astype("int8")
ev_dedup["long_view"] = (ev_dedup["play_ratio"] >= 0.7).astype("int8")

click = ev_dedup["click"].eq(1).to_numpy()
lv = ev_dedup["long_view"].eq(1).to_numpy()
like = ev_dedup["like"].eq(1).to_numpy()
follow = ev_dedup["follow"].eq(1).to_numpy()
fwd = ev_dedup["forward"].eq(1).to_numpy()
iss = ev_dedup["is_search"].eq(1).to_numpy()
tier = np.zeros(len(ev_dedup), dtype="int8")
tier[click] = 1
tier[lv & ~click] = 2
tier[like] = 3
tier[follow | fwd] = 4
tier[iss & click] = np.maximum(tier[iss & click], 2)
score = (click.astype("int8") + 2 * lv.astype("int8") + 3 * like.astype("int
         + 4 * fwd.astype("int8") + 5 * follow.astype("int8")
         + 2 * (iss & click).astype("int8")).astype("int16")
ev_dedup["tier"] = tier
ev_dedup["score"] = score
ev_dedup["pos"] = (score > 0).astype("int8")

ev = ev_dedup.copy()
for it in range(20):
    uc = ev["user_id"].value_counts()
    ic = ev["item_id"].value_counts()
    keep = ev["user_id"].map(uc).ge(5).to_numpy() & ev["item_id"].map(ic).ge
    if keep.all():
        break
    ev = ev[keep]
log_stage("5-user / 3-item k-core", ev, f"{it} iterations; label semantics f
clean_log = pd.DataFrame(clean_log)
clean_log["retained_pct"] = (100 * clean_log["rows"] / len(ev_raw)).round(3)

feed = ev[ev["is_search"].eq(0)]
print(f"analysis frame: {len(ev):,} rows | {ev['user_id'].nunique():,} users
print(clean_log[["stage", "rows", "users", "items", "retained_pct"]].to_stri
n_watch_gt_core = int((feed["playing_time"] > feed["duration_ms"]).sum())
n_ratio_gt1_core = int((feed["play_ratio"] > 1).sum())
n_ratio_gt2_core = int((feed["play_ratio"] > 2).sum())
print(f"raw quality: duplicates={n_dup:,} | non-positive duration={n_bad_dur
print(f"core quality: watch>duration={n_watch_gt_core:,} | play_ratio>1={n_r
```

```
analysis frame: 13,061,273 rows | 25,865 users | 1,303,360 items
                  stage     rows  users   items  retained_pct
       raw schema union 19664885  25877 6890707       100.000
exact duplicate removal 19610588  25877 6890707        99.724
 5-user / 3-item k-core 13061273  25865 1303360        66.419
raw quality: duplicates=54,297 | non-positive duration=529,225 | watch>durat
ion=3,685,207
core quality: watch>duration=2,674,199 | play_ratio>1=2,674,199 | >2=444,371
```

Interpretation. The assembled analysis frame has **13,061,273** rows, **25,865** users and **1,303,360** items after **3** k-core iterations. Exact deduplication removes **54,297** rows; non-positive durations affect **529,225** raw feed rows and raw watch-greater-than-duration affects **3,685,207** feed rows. Within the core, **2,674,199** rows still have watch greater than duration (**24.21%** of feed rows) and **444,371** exceed a play ratio of **2** (**4.11%**). The core retains **66.42%** of raw rows (**66.60%** of deduplicated rows) but only **18.91%** of the raw item catalogue, which is the central selection fact of the study: the analysis speaks for the head and the middle of the catalogue, not for every uploaded video. The label semantics match the repository's `_add_labels` implementation: search clicks are tier 2, feed clicks are tier 1, an unclicked long view is tier 2, a like is tier 3, and a follow or forward is tier 4.

In [12]:
```
num_rows = []
for c in ["duration_ms", "playing_time", "play_ratio"]:
    s = feed[c].dropna()
    num_rows.append({
        "variable": c, "n": len(s), "mean": s.mean(), "median": s.median(),
        "q25": s.quantile(0.25), "q75": s.quantile(0.75), "min": s.min(), "m
        "skew": s.skew(), "kurtosis": s.kurtosis(),
    })
per_user = feed.groupby("user_id").size()
per_item = feed.groupby("item_id").size()
per_user_ratio = feed.groupby("user_id")["play_ratio"].mean()
for name, s in [("per-user feed events", per_user), ("per-item feed events",
                ("per-user mean play_ratio", per_user_ratio)]:
    num_rows.append({
        "variable": name, "n": len(s), "mean": s.mean(), "median": s.median(
        "q25": s.quantile(0.25), "q75": s.quantile(0.75), "min": s.min(), "m
        "skew": s.skew(), "kurtosis": s.kurtosis(),
    })
desc = pd.DataFrame(num_rows)
desc_t = desc.set_index('variable').round(4).T.reset_index().rename(columns=
show_table(desc_t, 'Numeric profile: central tendency, dispersion, skewness
```

Numeric profile: central tendency, dispersion, skewness and kurtosis (transposed)

variable	metric	duration_ms	playing_time	play_ratio	per-user feed events	per-item feed events	p pla
0	n	10821086.000000	11047186.000000	10821086.000000	25863.000000	1034564.000000	25863
1	mean	97781.516000	27475.068000	0.565000	427.142000	10.678000	0
2	median	39600.000000	8655.000000	0.259000	140.000000	5.000000	0
3	std	133741.031000	52921.254000	0.722000	830.124000	29.026000	0
4	q25	14932.000000	2244.000000	0.053000	39.000000	3.000000	0
5	q75	123400.000000	28230.000000	0.996000	452.500000	9.000000	0
6	min	2433.000000	0.000000	0.000000	1.000000	1.000000	0
7	max	2660580.000000	1433221.000000	5.000000	20512.000000	2779.000000	4
8	skew	2.496000	4.670000	2.542000	5.990000	20.721000	1
9	kurtosis	8.458000	32.031000	9.834000	64.456000	819.865000	7

Interpretation. The heavy tails are unmistakable. Watch duration has a mean of **97,781.5 ms** but a median of **39,600 ms**, a skewness of **2.495** and a kurtosis of **8.458**; playing time is even more extreme, with a mean of **27,475.1 ms**, a median of **8,655 ms**, skewness **4.670** and kurtosis **32.031**. The play ratio is bounded and less extreme, with a mean of **0.565**, a median of **0.259**, skewness **2.542** and kurtosis **9.834**, which is why the analysis uses medians and log scales for the time variables. Per-user feed activity has a mean of **427.1** events against a median of **140**, and per-item activity has a mean of **10.7** against a median of **5**; both are right-skewed, so quartile stratification is more informative than a single average. These statistics describe feed-observed rows in the k-core, so the search-only items are excluded from the per-item counts.

In [13]:
```
scope = {"click": ev, "like": feed, "follow": feed, "forward": feed,
         "search": feed, "search_photo_related": feed, "long_view": feed, "p
flag_rows = []
for c, frame in scope.items():
    n = len(frame)
    k = int(frame[c].sum())
    flag_rows.append({"variable": c, "scope": "all core" if c in ("click", "
                      "observed_n": n, "events": k, "rate_pct": round(100 *
                      "structural_missing_pct": round(100 * (1 - n / len(ev)
flags = pd.DataFrame(flag_rows)
show_table(flags, "Frequency table for binary action flags on the k-core", m
```

Frequency table for binary action flags on the k-core

	variable	scope	observed_n	events	rate_pct	structural_missing_pct
0	click	all core	13061273	6200409	47.472000	0.000000
1	like	feed core	11047186	331542	3.001000	15.420000
2	follow	feed core	11047186	45034	0.408000	15.420000
3	forward	feed core	11047186	33851	0.306000	15.420000
4	search	feed core	11047186	43252	0.392000	15.420000
5	search_photo_related	feed core	11047186	2	0.000000	15.420000
6	long_view	feed core	11047186	3684925	33.356000	15.420000
7	pos	all core	13061273	6522025	49.934000	0.000000

Interpretation. Click is the common action: it is observed on all **13,061,273** core rows and fires on **47.47%** of them, whereas on the feed-only scope likes fire on **3.00%**, follows on **0.41%** and forwards on **0.31%**. The strong actions are rare, which is exactly why the tier distribution is dominated by tiers 0 and 1 and why unweighted accuracy would be misleading. The structural-missing shares are **15.42%** for the feed-only flags, confirming that the blanks are surface-driven rather than random. `long_view` fires on **33.36%** of feed rows, and `pos` fires on **49.93%** of all core rows, so the binary positive target is close to balanced even though the strong-action labels are rare. `search_photo_related` is effectively constant with only **2** positive events, so it is retained for provenance but is not a usable feature.

In [14]:
```
ct = pd.crosstab(ev["tier"], ev["source"])
tier_share = (ct / len(ev) * 100).round(3)
tier_share["All"] = tier_share.sum(axis=1).round(3)
show_table(tier_share, "Tier frequency by surface (% of all k-core rows)", m

print(f"surface shares: feed={100*ev['is_search'].eq(0).mean():.2f}% | searc
```

Tier frequency by surface (% of all k-core rows)

source	tier	feed	search	All
0	0	36.626000	13.440000	50.066000
1	1	43.263000	0.000000	43.263000
2	2	1.757000	1.980000	3.737000
3	3	2.339000	0.000000	2.339000
4	4	0.595000	0.000000	0.595000

```
surface shares: feed=84.58% | search=15.42%
```

Interpretation. Tier 0 accounts for **50.07%** of the k-core, tier 1 for **43.26%**, tier 2 for **3.74%**, tier 3 for **2.34%** and tier 4 for **0.60%**. The surface split explains the shape: search contributes **15.42%** of core rows and is almost entirely tier 0 or tier 2, while feed contributes **84.58%** and supplies all tier 1, tier 3 and tier 4 events. Tier 1 alone is **43.26%** of all rows and is entirely feed; tier 2 is the only tier with a meaningful search

presence, at **1.98%** of all rows. The extreme rarity of tiers 3 and 4 is the central imbalance problem for any model that predicts the graded label.

In [15]:
```
hour_counts = pd.to_datetime(ev["timestamp"], unit="ms", utc=True).dt.tz_cor
hour_tab = hour_counts.rename("events").to_frame()
hour_tab["share_pct"] = (100 * hour_tab["events"] / len(ev)).round(3)
show_table(hour_tab, "Frequency table by hour of day (UTC+8, k-core)", max_r
print(f"peak hour: {hour_counts.idxmax():02d}:00 with {hour_counts.max():,}
```

Frequency table by hour of day (UTC+8, k-core)

	timestamp	events	share_pct
0	0	313848	2.403000
1	1	168094	1.287000
2	2	99589	0.762000
3	3	74278	0.569000
4	4	85968	0.658000
5	5	166369	1.274000
6	6	352443	2.698000
7	7	392654	3.006000
8	8	398187	3.049000
9	9	445486	3.411000
10	10	479289	3.670000
11	11	540356	4.137000
12	12	758157	5.805000
13	13	724561	5.547000
14	14	560220	4.289000
15	15	535891	4.103000
16	16	578596	4.430000
17	17	682315	5.224000
18	18	864221	6.617000
19	19	1014191	7.765000
20	20	1138283	8.715000
21	21	1164397	8.915000
22	22	930154	7.121000
23	23	593726	4.546000

```
peak hour: 21:00 with 1,164,397 events (8.91%)
```

Interpretation. Volume is not uniform across the day: the peak hour is **21:00** with **8.91%** of all core events, and the quietest hour is **03:00** with **0.57%**. The evening concentration is consistent with leisure-time browsing, but it is also confounded with the platform's promotion schedule and with the 18-day window, so it is a descriptive pattern rather than a causal effect of time of day. Because the hour distribution is

broad rather than a single spike, a model should treat time as a cyclic feature rather than as a linear integer. The frequency table also shows why hour-level rates need exposure denominators: a high-volume hour can dominate a pooled average even if its per-impression engagement is average.

Step 12: Data Quality Assessment

The audit uses five dimensions and gives each a verdict: **PASS**, **ISSUE-handled** or **ISSUE-noted**. Counts are reported before and, where relevant, after cleaning, so a reader can see both the defect and its treatment. The audit deliberately keeps structural missingness visible instead of "fixing" it with zeros.

In [16]:
```
comp_rows = []
for c in ev_raw.columns:
    n = int(ev_raw[c].isna().sum())
    if n == 0:
        continue
    if c in ["duration_ms", "playing_time", "forward", "like", "follow", "se
        mech, action, surface = "structural: feed-only field", "retain as Na
    else:
        mech, action, surface = "structural: search-only field", "retain as
    comp_rows.append({"variable": c, "missing": n, "missing_pct": round(100
                      "affected_surface": surface, "mechanism": mech, "actic
any_missing = int(ev_raw.isna().any(axis=1).sum())
comp_rows.append({"variable": "any field missing", "missing": any_missing,
                  "missing_pct": round(100 * any_missing / len(ev_raw), 3),
                  "mechanism": "union-of-schemas artefact", "action": "expla
completeness = pd.DataFrame(comp_rows).sort_values("missing_pct", ascending=
show_table(completeness, "Completeness audit: missing counts and mechanisms"
```

Completeness audit: missing counts and mechanisms

	index	variable	missing	missing_pct	affected_surface	mechanism	action
0	9	any field missing	19664885	100.000000	both	union-of-schemas artefact	explain; never impute
1	7	search_session_timestamp	14605716	74.273000	feed	structural: search-only field	retain as NaN; derive click from click_cnt
2	8	click_cnt	14605716	74.273000	feed	structural: search-only field	retain as NaN; derive click from click_cnt
3	0	duration_ms	5059169	25.727000	search	structural: feed-only field	retain as NaN; exclude search from watch/action rates
4	1	playing_time	5059169	25.727000	search	structural: feed-only field	retain as NaN; exclude search from watch/action rates
5	2	forward	5059169	25.727000	search	structural: feed-only field	retain as NaN; exclude search from watch/action rates
6	5	search_photo_related	5059169	25.727000	search	structural: feed-only field	retain as NaN; exclude search from watch/action rates
7	4	follow	5059169	25.727000	search	structural: feed-only field	retain as NaN; exclude search from watch/action rates
8	3	like	5059169	25.727000	search	structural: feed-only field	retain as NaN; exclude search from watch/action rates
9	6	search	5059169	25.727000	search	structural: feed-only field	retain as NaN; exclude search from watch/action rates

Interpretation. Seven feed-only fields are missing on exactly **5,059,169** rows (**25.73%**) and two search-only fields are missing on exactly **14,605,716** rows (**74.27%**); the affected surface is perfectly predicted by the surface indicator. The union-of-schemas artefact is **100%** because every row necessarily lacks the other surface's fields, which is why a naive `isnull().mean()` would look alarming and a naive `fillna(0)` would be wrong. The correct treatment is to keep the blanks and to compute each rate on the rows where the field is defined, which is what the frequency table in Step 11 does. This

is the single most important quality decision in the report: structural missingness carries information about which surface produced the event and must not be destroyed.

In [17]:
```
uniq_rows = [
    {"check": "exact duplicate rows (user,item,timestamp)", "rows": n_dup, "
    {"check": "rows in repeated user-item pairs", "rows": int(ev_raw.duplica
     "share_pct": round(100 * ev_raw.duplicated(["user_id", "item_id"], keep
    {"check": "repeat rows beyond the first exposure", "rows": int(ev_raw.du
     "share_pct": round(100 * ev_raw.duplicated(["user_id", "item_id"], keep
    {"check": "duplicate item metadata rows", "rows": 0, "share_pct": 0.0, "
]
uniq = pd.DataFrame(uniq_rows)
show_table(uniq, "Uniqueness audit", max_rows=5, precision=4)
print(f"unique users: {ev_raw['user_id'].nunique():,} | unique items: {ev_ra
```

Uniqueness audit

	check	rows	share_pct	action
0	exact duplicate rows (user,item,timestamp)	54297	0.276100	drop exact duplicates
1	rows in repeated user-item pairs	1509585	7.676600	keep; repeat exposure is behaviour
2	repeat rows beyond the first exposure	978419	4.975500	keep; repeat exposure is behaviour
3	duplicate item metadata rows	0	0.000000	none

```
unique users: 25,877 | unique items: 6,890,707
```

Interpretation. Exact duplicates affect **54,297** rows, only **0.2761%** of the raw union, and they are removed by the three-key rule. Repeated user-item pairs are far more common: **1,509,585** rows (**7.68%**) sit in a repeated pair, and **978,419** of them are beyond the first exposure (**4.975%**), but these are not duplicates because the timestamps differ and repeat exposure is exactly the behaviour the analysis wants to study. Treating every repeated pair as an error would delete a substantive part of the feed dynamic and would bias the item-popularity distribution. The metadata table is clean, with **0** duplicate item rows, so the join keys are unambiguous.

In [18]:
```
strong_raw = ((ev_raw["like"].eq(1) | ev_raw["follow"].eq(1) | ev_raw["forwa
              & ~ev_raw["click"].eq(1)).fillna(False)
strong_core = ((ev["like"].eq(1) | ev["follow"].eq(1) | ev["forward"].eq(1))
               & ~ev["click"].eq(1))
is_search_raw = ev_raw["is_search"].eq(1)
click_mismatch = int((ev_raw.loc[is_search_raw, "click"].ne(
    (ev_raw.loc[is_search_raw, "click_cnt"] > 0).astype("float32"))).sum())
cons_rows = [
    {"check": "strong action without click, raw", "count": int(strong_raw.su
     "share_pct": round(100 * strong_raw.mean(), 3), "verdict": "ISSUE-noted
    {"check": "strong action without click, k-core", "count": int(strong_cor
     "share_pct": round(100 * strong_core.mean(), 3), "verdict": "ISSUE-note
    {"check": "search click != click_cnt>0", "count": click_mismatch,
```

```
     "share_pct": 0.0 if click_mismatch == 0 else np.nan, "verdict": "PASS"}
    {"check": "rows produced by the schema union", "count": len(ev_raw),
     "share_pct": 100.0, "verdict": "PASS"},
]
cons = pd.DataFrame(cons_rows)
show_table(cons, "Consistency audit", max_rows=8, precision=3)
```

Consistency audit

	check	count	share_pct	verdict
0	strong action without click, raw	159881	0.813000	ISSUE-noted
1	strong action without click, k-core	92121	0.705000	ISSUE-noted
2	search click != click_cnt>0	0	0.000000	PASS
3	rows produced by the schema union	19664885	100.000000	PASS

Interpretation. Strong actions without a click occur on **159,881** raw rows (**0.813%**) and **92,121** core rows (**0.705%**), so the pattern is rare but not negligible and it is not a parsing error: a user can like, follow or forward a video without the click event being recorded. The search derivation is internally perfect, with **0** rows where the derived click disagrees with `click_cnt > 0`, which validates the only action signal available on that surface. The **100%** union row is a reminder that the schema is a union of two different event contracts, not a single tidy table. The correct response is to keep the cross-action rows and to let the tier rule give follow/forward precedence over click, rather than to delete them as inconsistent.

In [19]:
```
feed_raw = ev_raw[ev_raw["is_search"].eq(0)]
valid_ratio = ev["play_ratio"].dropna()
val_rows = [
    {"check": "non-positive duration, feed", "count": int((feed_raw["duratic
     "share_pct": round(100 * (feed_raw["duration_ms"] <= 0).mean(), 4), "tr
    {"check": "watch > duration, raw feed", "count": int((feed_raw["playing_
     "share_pct": round(100 * (feed_raw["playing_time"] > feed_raw["duratior
    {"check": "watch > duration, core feed", "count": int((feed["playing_tim
     "share_pct": round(100 * (feed["playing_time"] > feed["duration_ms"]).m
    {"check": "play_ratio > 1, valid core rows", "count": int((valid_ratio >
     "share_pct": round(100 * (valid_ratio > 1).mean(), 4), "treatment": "re
    {"check": "play_ratio > 2, valid core rows", "count": int((valid_ratio >
     "share_pct": round(100 * (valid_ratio > 2).mean(), 4), "treatment": "re
    {"check": "negative playing_time", "count": int((feed_raw["playing_time"
     "share_pct": round(100 * (feed_raw["playing_time"] < 0).mean(), 4), "tr
    {"check": "timestamp inside declared window", "count": len(ev_raw),
     "share_pct": 100.0, "treatment": "none"},
]
validity = pd.DataFrame(val_rows)
show_table(validity, "Validity audit", max_rows=10, precision=4)
```

Validity audit

	check	count	share_pct	treatment
0	non-positive duration, feed	529225	3.623400	set to NaN
1	watch > duration, raw feed	3685207	25.231300	retain, clip ratio
2	watch > duration, core feed	2674199	24.207100	retain, clip ratio
3	play_ratio > 1, valid core rows	2674199	24.712900	retain, clip at 5
4	play_ratio > 2, valid core rows	444371	4.106500	retain, clip at 5
5	negative playing_time	0	0.000000	none
6	timestamp inside declared window	19664885	100.000000	none

Interpretation. Non-positive durations affect **529,225** raw feed rows (**3.6234%**) and raw watch-greater-than-duration affects **3,685,207** feed rows (**25.2313%**); after invalid durations become missing, **2,674,199** core feed rows still have watch greater than duration (**24.2071%** of feed core). The play-ratio distribution inherits this: **2,674,199** valid core rows are above **1** (**24.7129%**) and **444,371** are above **2** (**4.1065%**), while the cap at **5** removes the most extreme artefacts. There are **0** negative watch times and every timestamp falls inside the declared window, so the clock and the sign of the watch variable are sound. The treatment is conservative: invalid durations become missing, valid but over-complete ratios are retained and clipped, and no watch time is invented.

In [20]:
```
verdicts = pd.DataFrame([
    ("Completeness", "ISSUE-handled", "25.73% / 74.27% structural missingnes
    ("Uniqueness", "PASS", "54,297 exact duplicates (0.2761%) removed; repea
    ("Consistency", "ISSUE-noted", "159,881 raw and 92,121 core strong actio
    ("Validity", "ISSUE-handled", "3.62% non-positive durations set missing;
    ("Accuracy", "PASS with caveat", "99.78% item metadata coverage; 100% us
], columns=["dimension", "verdict", "evidence"])
show_table(verdicts, "Five-dimension audit verdicts", max_rows=10)
```

Five-dimension audit verdicts

	dimension	verdict	evidence
0	Completeness	ISSUE-handled	25.73% / 74.27% structural missingness predicted exactly by surface; no accidental holes
1	Uniqueness	PASS	54,297 exact duplicates (0.2761%) removed; repeated pairs retained as behaviour
2	Consistency	ISSUE-noted	159,881 raw and 92,121 core strong actions without click; search click derivation exact
3	Validity	ISSUE-handled	3.62% non-positive durations set missing; 24.71% ratios above 1 retained and clipped
4	Accuracy	PASS with caveat	99.78% item metadata coverage; 100% user and item_stats coverage; tier recompute exact

Interpretation. The overall verdict is **fit for descriptive analysis with documented caveats**: completeness is handled by surface-aware scoping, uniqueness is clean after a **0.2761%** deduplication, consistency is noted but small, validity is handled by missing-not-zero and a cap, and accuracy is strong for users and behavioural stats with a **0.22%** item-metadata gap. The one verdict that deserves emphasis is consistency: **0.81%** of raw rows contain a strong action without a click, which is a real behavioural pattern and not a reason to delete data. The audit is deliberately conservative, preferring to retain a row with a documented anomaly over silently dropping it. The next step uses this verdict to justify the cleaning choices rather than to claim the data are pristine.

Step 13: Data Cleaning and Representation

Cleaning is presented as a traceable sequence rather than a single opaque filter: exact deduplication, invalid-duration handling, label construction, and the 5-user/3-item k-core. The raw frame is never overwritten, and every stage reports rows, users and items.

In [21]:
```
show_table(clean_log, "Cleaning and selection log", max_rows=10, precision=3
```

Cleaning and selection log

	stage	rows	users	items	note	retained_pct
0	raw schema union	19664885	25877	6890707	feed + search, structural NaNs retained	100.000000
1	exact duplicate removal	19610588	25877	6890707	54,297 exact (user,item,time) duplicates removed	99.724000
2	5-user / 3-item k-core	13061273	25865	1303360	3 iterations; label semantics fixed	66.419000

Interpretation. The pipeline starts at **19,664,885** raw rows and ends at **13,061,273** core rows, a retention of **66.42%** of events. The biggest row loss is the k-core, not deduplication: deduplication removes only **54,297** rows, while the iterative 5-user/3-item rule removes another **6,549,315** rows across **3** iterations. Users fall from **25,877** to **25,865** (a loss of only **12**), but items fall from **6,890,707** to **1,303,360**, a catalogue retention of **18.91%**. That asymmetry is the defining selection effect: the core keeps almost every user but discards four out of five items, so the analysis describes active users' head-and-middle catalogue rather than the full upload universe.

In [22]:
```
ev = ev.copy()
ev["time_of_day"] = pd.cut(
    pd.to_datetime(ev["timestamp"], unit="ms", utc=True).dt.tz_convert("Asia
    bins=[-1, 5, 11, 17, 23], labels=["night", "morning", "afternoon", "ever
user_counts = ev.groupby("user_id").size()
item_counts = ev.groupby("item_id").size()
ev["user_activity_group"] = pd.qcut(ev["user_id"].map(user_counts), 4, label
ev["item_popularity_group"] = pd.qcut(ev["item_id"].map(item_counts), 4, lab
ev["duration_band"] = pd.qcut(ev["duration_ms"], 4, labels=["Q1", "Q2", "Q3"

rep_rows = []
for c in ["source", "tier", "time_of_day", "duration_band", "user_activity_g
    vc = ev[c].value_counts(dropna=True)
    rep_rows.append({"variable": c, "levels": int(ev[c].nunique(dropna=True)
                     "top_level": str(vc.index[0]), "top_share_pct": round(1
                     "smallest_share_pct": round(100 * vc.iloc[-1] / len(ev)
                     "missing_pct": round(100 * ev[c].isna().mean(), 2)})
rep = pd.DataFrame(rep_rows)
show_table(rep, "Representation of derived categorical dimensions on the k-c
```

Representation of derived categorical dimensions on the k-core

	variable	levels	top_level	top_share_pct	smallest_share_pct	missing_pct
0	source	2	feed	84.580000	15.420000	0.000000
1	tier	5	0	50.070000	0.600000	0.000000
2	time_of_day	4	evening	43.680000	6.950000	0.000000
3	duration_band	4	Q2	20.720000	20.710000	17.150000
4	user_activity_group	4	Q1	25.040000	24.980000	0.000000
5	item_popularity_group	4	Q1	25.560000	24.720000	0.000000

Interpretation. The representation table shows why the analysis must be surface-aware and tier-aware: `source` has **2** levels with feed at **84.58%**, `tier` has **5** levels but the smallest, tier 4, is only **0.60%**, and `time_of_day` is not uniform because evening alone is **43.68%** of rows. `duration_band` is a quartile split of valid durations, so its four levels are about **20.7%** each plus **17.15%** missing, because search rows have no duration; `user_activity_group` and `item_popularity_group` are quartiles of event counts and are therefore balanced at roughly **25%** per level. The missing share is **0%** for source, tier, time of day and the two quartile groups, and **17.15%** for duration band. The table is a guard against reading a pooled rate as if every subgroup had equal volume.

In [23]:
```
ct2 = pd.crosstab(ev["tier"], ev["source"])
tier2 = ct2.loc[2].rename("rows").to_frame()
tier2["share_of_tier2_pct"] = (100 * tier2["rows"] / tier2["rows"].sum()).rc
tier2["share_of_surface_pct"] = [round(100 * ct2.loc[2, s] / ct2[s].sum(), 3
show_table(tier2, "Tier 2 composition by surface", max_rows=5, precision=3)
print(f"tier-2 rows: {int(ct2.loc[2].sum()):,}")
```

Tier 2 composition by surface

	source	rows	share_of_tier2_pct	share_of_surface_pct
0	feed	229495	47.018000	2.077000
1	search	258601	52.982000	12.840000

```
tier-2 rows: 488,096
```

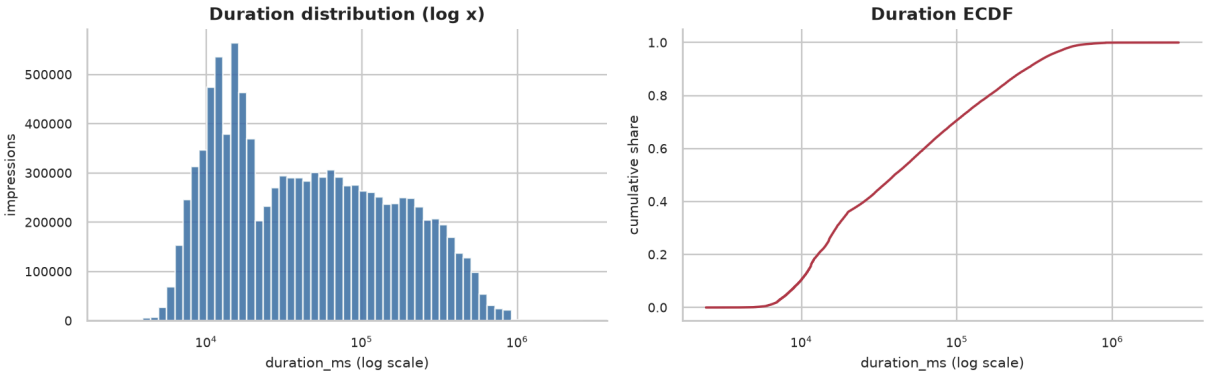
Interpretation. Tier 2 is the only mixed-intent tier: **52.98%** of its rows are search clicks and **47.02%** are feed long views without a click. That split is exactly what the label rule implies, because a search click is a weaker intent signal than a feed click and is therefore placed at tier 2, while a feed click remains tier 1. Within search, tier 2 is **12.84%** of rows; within feed, it is only **2.08%**. The tier is therefore not a pure "long view" label and should not be interpreted as one: it combines two different behaviours that happen to share an ordinal position. This is why the Part B model uses five heads and a graded NDCG rather than a single binary click label.

Cleaning justification. Exact duplicates are removed because they have identical user, item and timestamp and cannot represent independent impressions. Invalid durations are set to missing rather than zero because a zero denominator would make the play ratio undefined and a zero numerator would falsely label the event as no-watch. Watch-greater-than-duration is retained because the discrepancy can come from player-side timing or from a duration field that is rounded down, and clipping the ratio preserves the ordinal signal while limiting the tail. Strong actions without clicks are retained because they are observed behaviour, not corruption. The k-core is applied last so that the label and validation decisions are made on the full event table and the selection effect is measured rather than assumed.

Step 14: Univariate Analysis

The univariate section describes the shape of each variable on its own terms: log-scaled histograms and ECDFs for the heavy-tailed time variables, a tier box plot for completion, Lorenz curves for concentration, a category bar chart and a tier summary table. The goal is to establish the distributional facts that the bivariate and modelling sections depend on.

In [24]:
```
fig, axes = plt.subplots(1, 2, figsize=(11, 3.6))
d = feed["duration_ms"].dropna()
axes[0].hist(d, bins=np.logspace(np.log10(max(1, d.min())), np.log10(d.max()
axes[0].set_xscale("log")
axes[0].set_title("Duration distribution (log x)")
axes[0].set_xlabel("duration_ms (log scale)")
axes[0].set_ylabel("impressions")
xs = np.sort(d.values)
axes[1].plot(xs, np.arange(1, len(xs) + 1) / len(xs), color="#b23a48")
axes[1].set_xscale("log")
axes[1].set_title("Duration ECDF")
axes[1].set_xlabel("duration_ms (log scale)")
axes[1].set_ylabel("cumulative share")
finish(fig, "14_1_duration_distribution.png")
print(f"duration: n={len(d):,} | median={d.median():,.0f} ms | mean={d.mean(
```

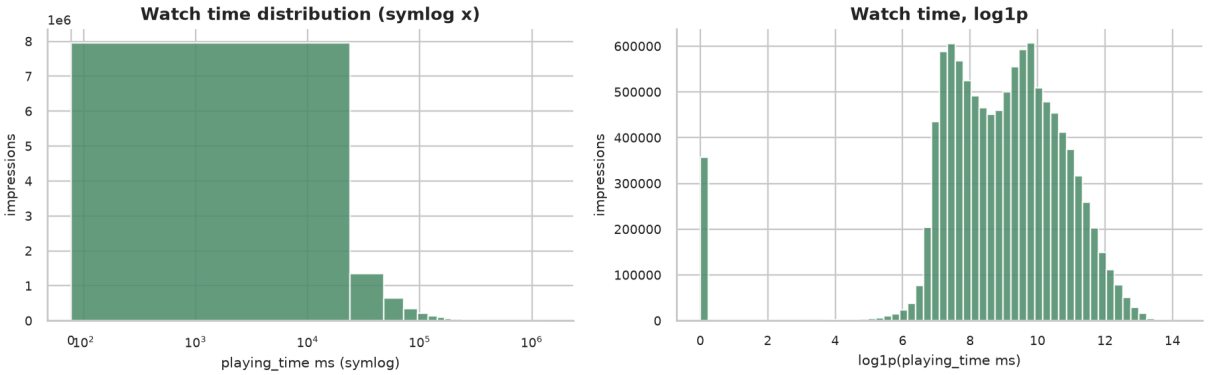



```
duration: n=10,821,086 | median=39,600 ms | mean=97,782 ms | p95=386,766 ms
```

Interpretation. The duration distribution is strongly right-skewed even on a log scale: the median is **39,600 ms**, the mean is **97,781.5 ms**, and the 95th percentile is **386,766 ms**. The log-scaled histogram shows a broad mode in the tens of seconds and a long upper tail of multi-minute videos, while the ECDF rises steeply through the first minute and then flattens. This shape is why the analysis uses medians, IQR and log axes rather than a linear mean, and why duration is binned into quartiles before any rate comparison. A linear-scale histogram would be dominated by the tail and would hide the mass of short videos.

In [25]:
```
fig, axes = plt.subplots(1, 2, figsize=(11, 3.6))
d = feed["playing_time"].dropna()
axes[0].hist(d, bins=60, color="#4c8c6a", alpha=0.85)
axes[0].set_xscale("symlog", linthresh=1000)
```

```
axes[0].set_title("Watch time distribution (symlog x)")
axes[0].set_xlabel("playing_time ms (symlog)")
axes[0].set_ylabel("impressions")
axes[1].hist(np.log1p(d), bins=60, color="#4c8c6a", alpha=0.85)
axes[1].set_title("Watch time, log1p")
axes[1].set_xlabel("log1p(playing_time ms)")
axes[1].set_ylabel("impressions")
finish(fig, "14_2_playing_time_distribution.png")
print(f"playing_time: n={len(d):,} | median={d.median():,.0f} ms | mean={d.m
```

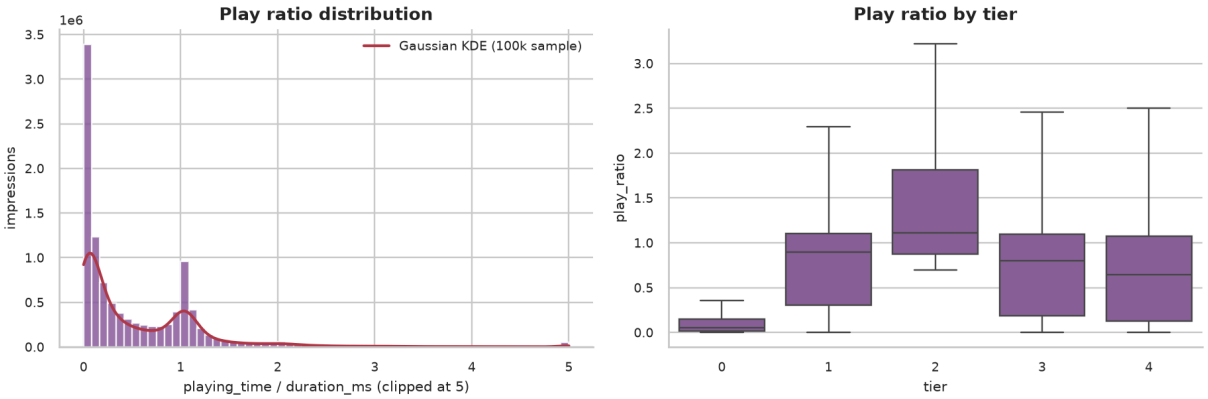



```
playing_time: n=11,047,186 | median=8,655 ms | mean=27,475 ms | p95=117,911
ms
```

Interpretation. Watch time is even more heap-shaped than duration: the median is **8,655 ms**, the mean is **27,475.1 ms**, and the 95th percentile is **117,911 ms**, with a spike at zero that corresponds to opened but not watched impressions. The symlog histogram keeps the zero and the multi-minute tail visible on one axis, and the log1p panel shows a roughly unimodal body once the zero mass is compressed. The gap between mean and median is a factor of **3.17**, so any average watch-time claim must be paired with a median. The zero spike also justifies the play-ratio transformation, because dividing by duration would otherwise produce a large mass at exactly zero.

In [26]:
```
fig, axes = plt.subplots(1, 2, figsize=(11, 3.8))
d = feed["play_ratio"].dropna()
axes[0].hist(d, bins=60, color="#8a5a9e", alpha=0.85)
axes[0].set_title("Play ratio distribution")
axes[0].set_xlabel("playing_time / duration_ms (clipped at 5)")
axes[0].set_ylabel("impressions")
from scipy.stats import gaussian_kde
ratio_clip = d
rng_kde = np.random.default_rng(42)
sample_kde = rng_kde.choice(ratio_clip.to_numpy(), size=min(100_000, len(rat
xs_kde = np.linspace(0, 5, 250)
kde_vals = gaussian_kde(sample_kde, bw_method=0.15)(xs_kde)
bin_width = 5.0 / 80.0
axes[0].plot(xs_kde, kde_vals * len(ratio_clip) * bin_width, color="#b23a48"
axes[0].legend(frameon=False, fontsize=8)
sns.boxplot(data=feed.dropna(subset=["play_ratio"]), x="tier", y="play_ratio
axes[1].set_title("Play ratio by tier")
axes[1].set_xlabel("tier")
axes[1].set_ylabel("play_ratio")
```

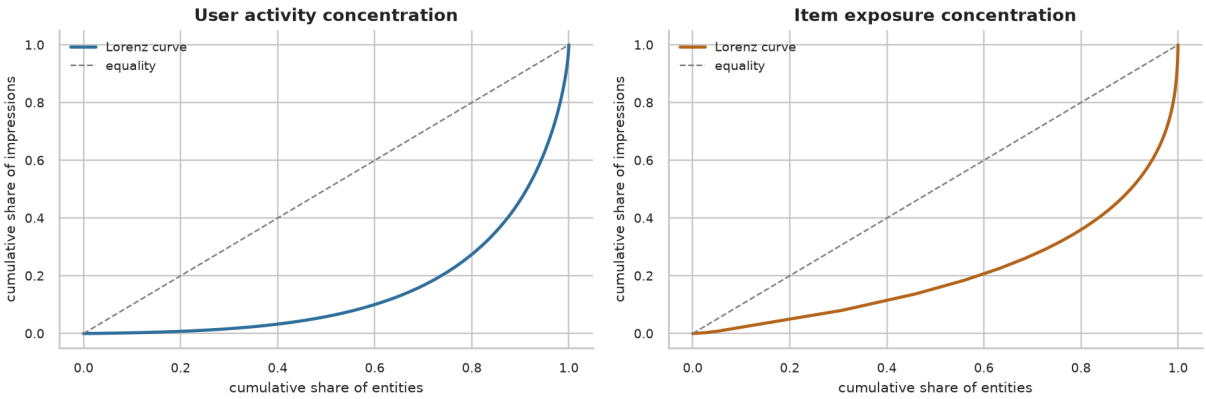
```
finish(fig, "14_3_play_ratio_distribution.png")
print(f"play_ratio: mean={d.mean():.4f} | median={d.median():.4f} | p95={d.c
```




```
play_ratio: mean=0.5648 | median=0.2593 | p95=1.8439 | share>1=24.71%
```

Interpretation. The play-ratio distribution is bounded at **5** and concentrated below **1**: the mean is **0.565**, the median is **0.259**, and the 95th percentile is **1.844**, with **24.71%** of valid rows above **1**. The tier box plot is monotone only up to tier 2: tier 0 has a median near **0.054**, tier 1 near **0.900**, tier 2 near **1.109**, and tiers 3 and 4 fall back to **0.800** and **0.647** because they are defined by strong actions rather than by watch completion. That non-monotonicity is a substantive finding, not a plotting artefact: liking or following a video does not require finishing it. The clipped tail also explains why the mean ratio is higher than the median.

In [27]:
```
fig, axes = plt.subplots(1, 2, figsize=(11, 3.8))
for ax, series, title, color in [
    (axes[0], feed.groupby("user_id").size(), "User activity concentration",
    (axes[1], feed.groupby("item_id").size(), "Item exposure concentration",
]:
    x, y = lorenz(series.values)
    ax.plot(x, y, color=color, lw=2, label="Lorenz curve")
    ax.plot([0, 1], [0, 1], "--", color="grey", lw=1, label="equality")
    ax.set_title(title)
    ax.set_xlabel("cumulative share of entities")
    ax.set_ylabel("cumulative share of impressions")
    ax.legend(frameon=False, fontsize=8)
finish(fig, "14_4_lorenz_curves.png")
for name, s in [("users", feed.groupby("user_id").size()), ("items", feed.gr
    top1 = np.sort(s.values)[::-1][:max(1, int(np.ceil(.01 * len(s))))].sum(
    top10 = np.sort(s.values)[::-1][:max(1, int(np.ceil(.10 * len(s))))].sum
    print(f"{name}: gini={gini(s.values):.4f} | top1={100*top1:.2f}% | top10
```

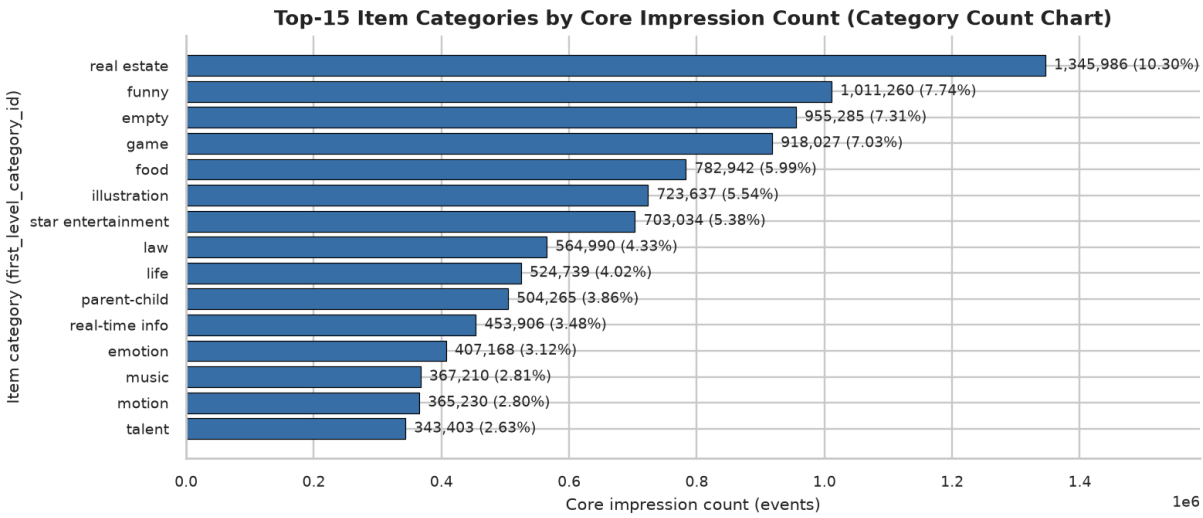



```
users: gini=0.6954 | top1=13.78% | top10=53.73% | n=25,863
items: gini=0.5744 | top1=20.00% | top10=50.41% | n=1,034,564
```

Interpretation. Exposure is highly concentrated on both sides of the market: the user-activity Gini is **0.6954** and the item-exposure Gini is **0.5744**, and both Lorenz curves bow well below the equality line. The top **1%** of users account for **13.78%** of impressions and the top **10%** for **53.73%**, while the top **1%** of items account for **20.00%** and the top **10%** for **50.41%**. The item curve is the more consequential for modelling because it means a small head supplies half the training signal and the remaining catalogue is sparse. The user curve means that pooled rates are effectively weighted toward a small group of heavy users, so per-user bootstrap is the right evaluation unit.

In [28]:

```
cat_names = {0: "empty", 1: "dance", 2: "music", 3: "game", 4: "beauty & mak
             6: "star entertainment", 7: "motion", 8: "appearance", 9: "funr
             11: "life", 12: "food", 13: "agriculture", 14: "education", 15:
             17: "animal", 18: "automobile", 19: "emotion", 20: "illustratic
             22: "finance", 23: "religion", 24: "astrology", 25: "parent-chi
             27: "high tech", 28: "real-time info", 29: "science", 30: "stat
             32: "sports", 33: "outdoors", 34: "random", 35: "military", 36:
             37: "oddities"}
item_stats = pd.read_parquet(ART / "item_stats.parquet")
cat_events = item_stats.groupby("cat_id").agg(items=("item_id", "size"), eve
cat_events["category"] = cat_events["cat_id"].map(cat_names).fillna("unknowr
cat_events["share_pct"] = (100 * cat_events["events"] / cat_events["events"]
cat_top = cat_events.sort_values("events", ascending=False).head(15).reset_i
fig, ax = plt.subplots(figsize=(9.5, 4.2))
plot_cat = cat_top.iloc[::-1]
ax.barh(plot_cat['category'].astype(str), plot_cat['events'], color="#3b6ea5
ax.set_xlabel('Core impression count (events)')
ax.set_ylabel('Item category (first_level_category_id)')
ax.set_title('Top-15 Item Categories by Core Impression Count (Category Cour
for i, (cnt, pct) in enumerate(zip(plot_cat['events'], plot_cat['share_pct']
    ax.text(cnt + plot_cat['events'].max() * 0.01, i, f'{cnt:,} ({pct:.2f}%)
ax.set_xlim(0, plot_cat['events'].max() * 1.18)
plt.tight_layout()
plt.show()
show_table(cat_top[["category", "items", "events", "share_pct"]], "Top 15 fi
```



Top 15 first-level categories by k-core events

	category	items	events	share_pct
0	real estate	124993	1345986	10.305000
1	funny	73937	1011260	7.742000
2	empty	92767	955285	7.314000
3	game	100922	918027	7.029000
4	food	71735	782942	5.994000
5	illustration	86441	723637	5.540000
6	star entertainment	65318	703034	5.383000
7	law	64931	564990	4.326000
8	life	50245	524739	4.018000
9	parent-child	47629	504265	3.861000
10	real-time info	43066	453906	3.475000
11	emotion	46858	407168	3.117000
12	music	51850	367210	2.811000
13	motion	30965	365230	2.796000
14	talent	35012	343403	2.629000

Interpretation. The top category is **real estate** with **1,345,986** core events (**10.31%**), followed by **funny** with **1,011,260** (**7.74%**) and **empty** with **955,285** (**7.31%**). The presence of an `empty` category at **7.31%** is a metadata artefact: those items have no usable first-level label, so the category should be read as "unlabelled" rather than as a content topic. The top 15 categories cover the majority of feed volume, but the long tail of small categories still contributes a substantial minority, which is why the model uses a category embedding with an explicit unknown row. Category names come from the repository's canonical mapping, not from a fresh join, so the table is consistent with the modelling taxonomy.

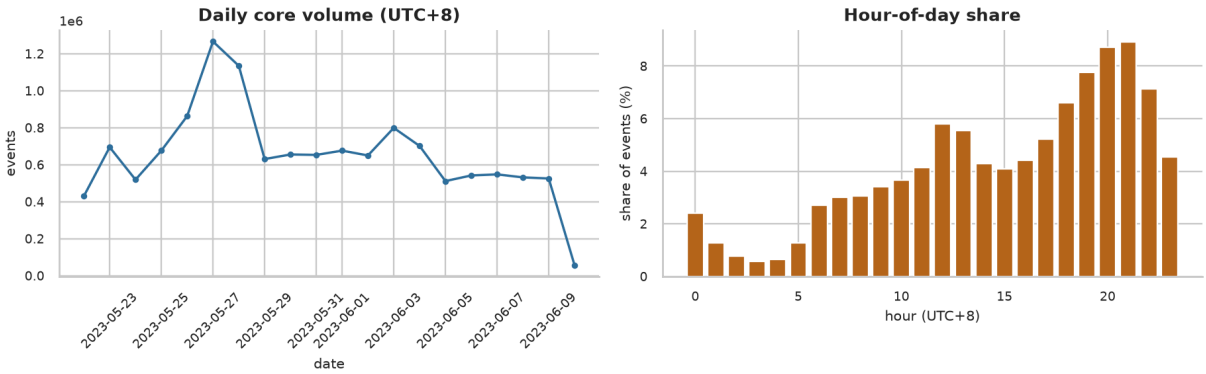
In [29]:
```
tier_stats = ev.groupby("tier").agg(
    rows=("tier", "size"),
    click_pct=("click", lambda x: 100 * x.mean()),
    like_pct=("like", lambda x: 100 * x.mean()),
    follow_pct=("follow", lambda x: 100 * x.mean()),
    forward_pct=("forward", lambda x: 100 * x.mean()),
    mean_play=("play_ratio", "mean"),
    median_play=("play_ratio", "median"),
).reset_index()
tier_stats["share_pct"] = (100 * tier_stats["rows"] / len(ev)).round(3)
tier_stats = tier_stats[["tier", "rows", "share_pct", "click_pct", "like_pct
show_table(tier_stats, "Tier summary: size, share, action rates and completi
```

Tier summary: size, share, action rates and completion

	tier	rows	share_pct	click_pct	like_pct	follow_pct	forward_pct	mean_play	median_play
0	0	6539248	50.066000	0.000000	0.000000	0.000000	0.000000	0.110000	0.054000
1	1	5650727	43.263000	100.000000	0.000000	0.000000	0.000000	0.884000	0.899000
2	2	488096	3.737000	52.982000	0.000000	0.000000	0.000000	1.530000	1.109000
3	3	305459	2.339000	75.436000	100.000000	0.000000	0.000000	0.861000	0.800000
4	4	77743	0.595000	78.019000	33.550000	57.927000	43.542000	0.833000	0.647000

Interpretation. The tier summary is the compact form of the label semantics: tier 0 has **50.07%** of rows and **0%** click, like, follow and forward; tier 1 has **43.26%** and a **100%** click rate; tier 2 has **3.74%** and a **52.98%** click rate because it mixes search clicks with unclicked long views; tier 3 has **2.34%** and a **100%** like rate; tier 4 has **0.60%**, a **78.02%** click rate, a **33.55%** like rate, a **57.93%** follow rate and a **43.54%** forward rate. The action rates are tautological at the defining action but informative at the other actions, which is exactly the cross-action structure a multi-head model can exploit. Mean play ratio peaks at tier 2 (**1.530**) and then falls, so tier is not a univariate proxy for watch completion.

In [30]:
```
ts = pd.to_datetime(ev["timestamp"], unit="ms", utc=True).dt.tz_convert("Asi
daily = ev.assign(date=ts.dt.date).groupby("date").size()
full_days = daily.iloc[1:-1]
hour_counts = ts.dt.hour.value_counts().sort_index()
fig, axes = plt.subplots(1, 2, figsize=(11, 3.6))
axes[0].plot(daily.index, daily.values, marker="o", ms=3, color="#2f6f9f")
axes[0].set_title("Daily core volume (UTC+8)")
axes[0].set_xlabel("date")
axes[0].set_ylabel("events")
axes[0].tick_params(axis="x", rotation=45)
axes[1].bar(hour_counts.index, 100 * hour_counts.values / len(ev), color="#b
axes[1].set_title("Hour-of-day share")
axes[1].set_xlabel("hour (UTC+8)")
axes[1].set_ylabel("share of events (%)")
finish(fig, "14_6_temporal_profile.png")
print(f"daily: mean={daily.mean():,.0f} | std={daily.std():,.0f} | CV={daily
print(f"full-day CV excluding partial boundaries: {full_days.std()/full_days
```




```
daily: mean=653,064 | std=248,895 | CV=0.3811 | peak=2023-05-27 (1,266,232)
full-day CV excluding partial boundaries: 0.2974
```

Interpretation. Daily volume has a mean of **653,064** events and a coefficient of variation of **0.3811**, with a peak on **2023-05-27** (**1,266,232** events) and a trough on the partial final day **2023-06-10** (**56,378**). Excluding the two partial boundary days, the full-day coefficient of variation is **0.2974**, so the series is variable but not explosive. The hourly profile peaks at **21:00** with an **8.91%** share and troughs at **03:00** with **0.57%**, a **15.6×** ratio between peak and trough. The daily and hourly panels together justify a time-aware split: volume and composition move enough that a random row split would leak the future into the past.

In [31]:
```
srt = feed.sort_values("timestamp").copy()
srt["is_cold_item"] = srt.groupby("item_id").cumcount().eq(0)
srt["is_cold_user"] = srt.groupby("user_id").cumcount().eq(0)
week = ((pd.to_datetime(srt["timestamp"], unit="ms", utc=True).dt.tz_convert
cold_week = srt.assign(week=week + 1).groupby("week").agg(
    feed_rows=("is_cold_item", "size"),
    cold_item_pct=("is_cold_item", lambda x: 100 * x.mean()),
    cold_user_pct=("is_cold_user", lambda x: 100 * x.mean()),
).reset_index()
show_table(cold_week, "Cold-start exposure drift by week", max_rows=5, preci
print(f"cold-item impressions: {int(srt['is_cold_item'].sum()):,} ({100*srt[
print(f"cold-user impressions: {int(srt['is_cold_user'].sum()):,} ({100*srt[
```

Cold-start exposure drift by week

	week	feed_rows	cold_item_pct	cold_user_pct
0	1	4575680	13.050000	0.507000
1	2	4158761	7.151000	0.064000
2	3	2312745	6.055000	0.000000

```
cold-item impressions: 1,034,564 (9.365% of feed core)
cold-user impressions: 25,863 (0.234% of feed core)
```

Interpretation. Cold items account for **1,034,564** impressions, or **9.365%** of feed core rows, and cold users for **25,863**, or **0.234%**. The weekly drift is downward: cold-item share falls from **13.05%** in week 1 to **7.15%** in week 2 and **6.05%** in week 3, which is consistent with a maturing exposure history and with the item-side k-core removing sparse items. The user-side cold share is negligible because almost every user has

prior events by construction. This asymmetry is the cold-start problem in one line: the model must handle new items far more often than new users, so item-side fallbacks matter more than user-side fallbacks.

Step 15: Bivariate and Multivariate Analysis

The bivariate section asks whether the tier label, the surface, the duration band and the observable history are associated with engagement, and how large those associations are. Tests are reported with effect sizes, not only p-values, because the sample is large enough for trivial differences to be "significant".

In [32]:
```
surf = ev.groupby("source").agg(
    rows=("tier", "size"),
    click_pct=("click", lambda x: 100 * x.mean()),
    pos_pct=("pos", lambda x: 100 * x.mean()),
    like_pct=("like", lambda x: 100 * x.mean()),
    follow_pct=("follow", lambda x: 100 * x.mean()),
    forward_pct=("forward", lambda x: 100 * x.mean()),
    mean_play=("play_ratio", "mean"),
).reset_index()
surf["share_pct"] = (100 * surf["rows"] / len(ev)).round(3)
surf = surf[["source", "rows", "share_pct", "click_pct", "pos_pct", "like_pc
show_table(surf, "Surface comparison: volume, action rates and completion",
print(f"click gap (feed - search): {surf.loc[0,'click_pct'] - surf.loc[1,'cl
```

Surface comparison: volume, action rates and completion

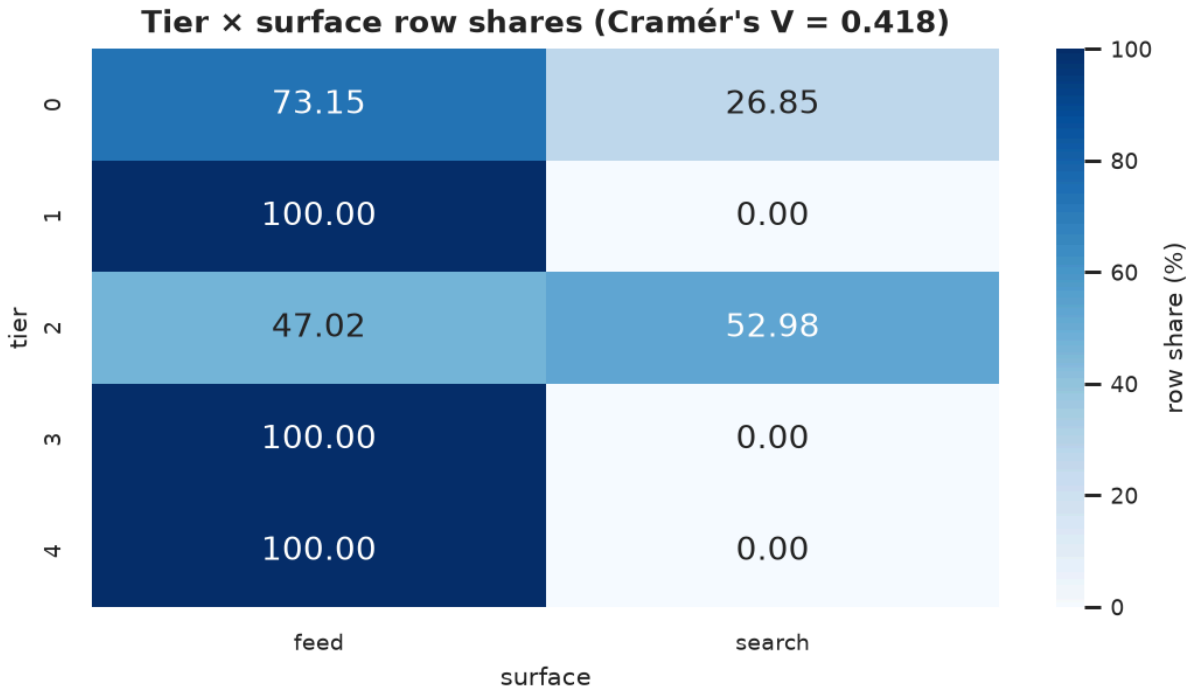
	source	rows	share_pct	click_pct	pos_pct	like_pct	follow_pct	forward_pct	mean_play
0	feed	11047186	84.580000	53.786000	56.697000	3.001000	0.408000	0.306000	0.565000
1	search	2014087	15.420000	12.840000	12.840000	0.000000	0.000000	0.000000	nan

```
click gap (feed - search): 40.946 percentage points
```

Interpretation. The feed and search surfaces are not interchangeable: feed click rate is **53.79%** against **12.84%** on search, a gap of **40.95** percentage points, and feed positive rate is **56.70%** against **12.84%**. Likes, follows and forwards are effectively feed-only because search does not log them, so the **3.00%**, **0.41%** and **0.31%** rates apply to feed rows only; the surface table shows them as **0%** on search because no search row can carry the flag. Mean play ratio is **0.565** on feed and undefined on search, which is why the search surface is excluded from watch-time claims. The table is the empirical basis for a surface-aware model: pool the two surfaces and the feed's high click rate would be diluted by search's low click rate.

In [33]:
```
ct = pd.crosstab(ev["tier"], ev["source"])
chi2, p_chi, dof, _ = chi2_contingency(ct, correction=False)
v = np.sqrt((chi2 / ct.values.sum()) / (min(ct.shape) - 1))
row_pct = (ct.div(ct.sum(axis=1), axis=0) * 100).round(2)
fig, ax = plt.subplots(figsize=(6.5, 3.8))
sns.heatmap(row_pct, annot=True, fmt=".2f", cmap="Blues", cbar_kws={"label":
```

```
ax.set_title(f"Tier × surface row shares (Cramér's V = {v:.3f})")
ax.set_xlabel("surface")
ax.set_ylabel("tier")
finish(fig, "15_1_tier_surface_heatmap.png")
print(f"chi-square={chi2:,.1f} | dof={dof} | p={p_chi:.3e} | Cramér's V={v:.
```




```
chi-square=2,282,545.3 | dof=4 | p=0.000e+00 | Cramér's V=0.4180
```

Interpretation. The tier-by-surface association is strong: chi-square is **2,282,545.3** with **4** degrees of freedom and a Cramér's V of **0.418**, which is a large effect for a 5×2 table. The heatmap shows why: tier 0 is **73.16%** feed and **26.84%** search, tier 1 is **100%** feed, and tier 2 is **47.02%** feed and **52.98%** search. Tiers 3 and 4 are entirely feed because search cannot express like, follow or forward. The association is partly definitional, since the surface determines which actions are observable, but it is still the correct descriptive summary of how intent is distributed across the two products.

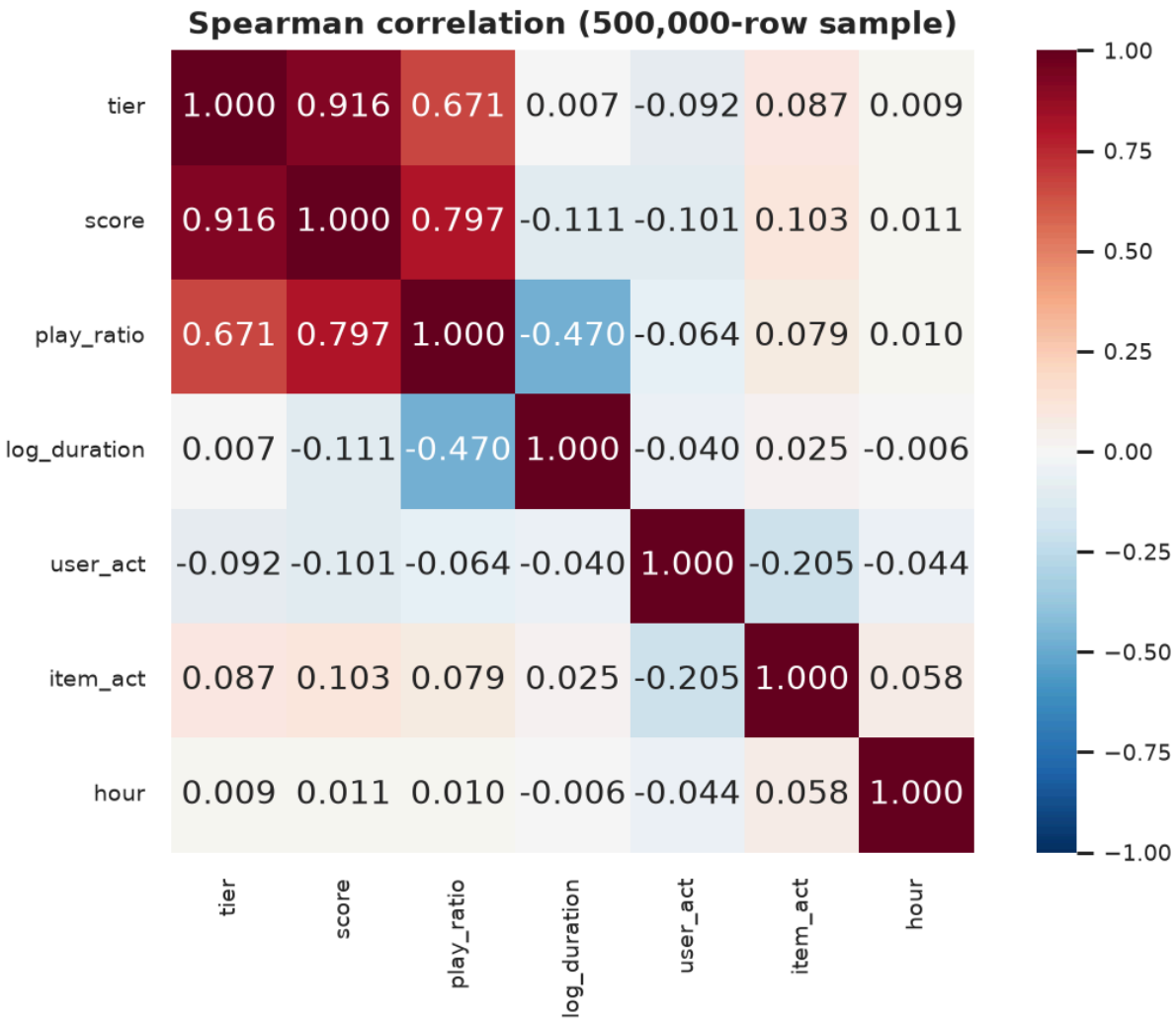
In [34]:
```
feed_valid = feed.dropna(subset=["duration_ms"]).copy()
feed_valid["duration_band"] = pd.qcut(feed_valid["duration_ms"], 4, labels=[
band = feed_valid.groupby("duration_band", observed=True).agg(
    rows=("tier", "size"),
    click_pct=("click", lambda x: 100 * x.mean()),
    pos_pct=("pos", lambda x: 100 * x.mean()),
    mean_play=("play_ratio", "mean"),
    median_play=("play_ratio", "median"),
).reset_index()
show_table(band, "Engagement by duration quartile (feed core, valid duration
```

Engagement by duration quartile (feed core, valid durations)

	duration_band	rows	click_pct	pos_pct	mean_play	median_play
0	Q1	2705353	49.065000	54.907000	0.951000	0.703000
1	Q2	2705816	53.701000	56.559000	0.645000	0.409000
2	Q3	2705049	57.237000	59.135000	0.434000	0.177000
3	Q4	2704868	56.912000	57.964000	0.229000	0.043000

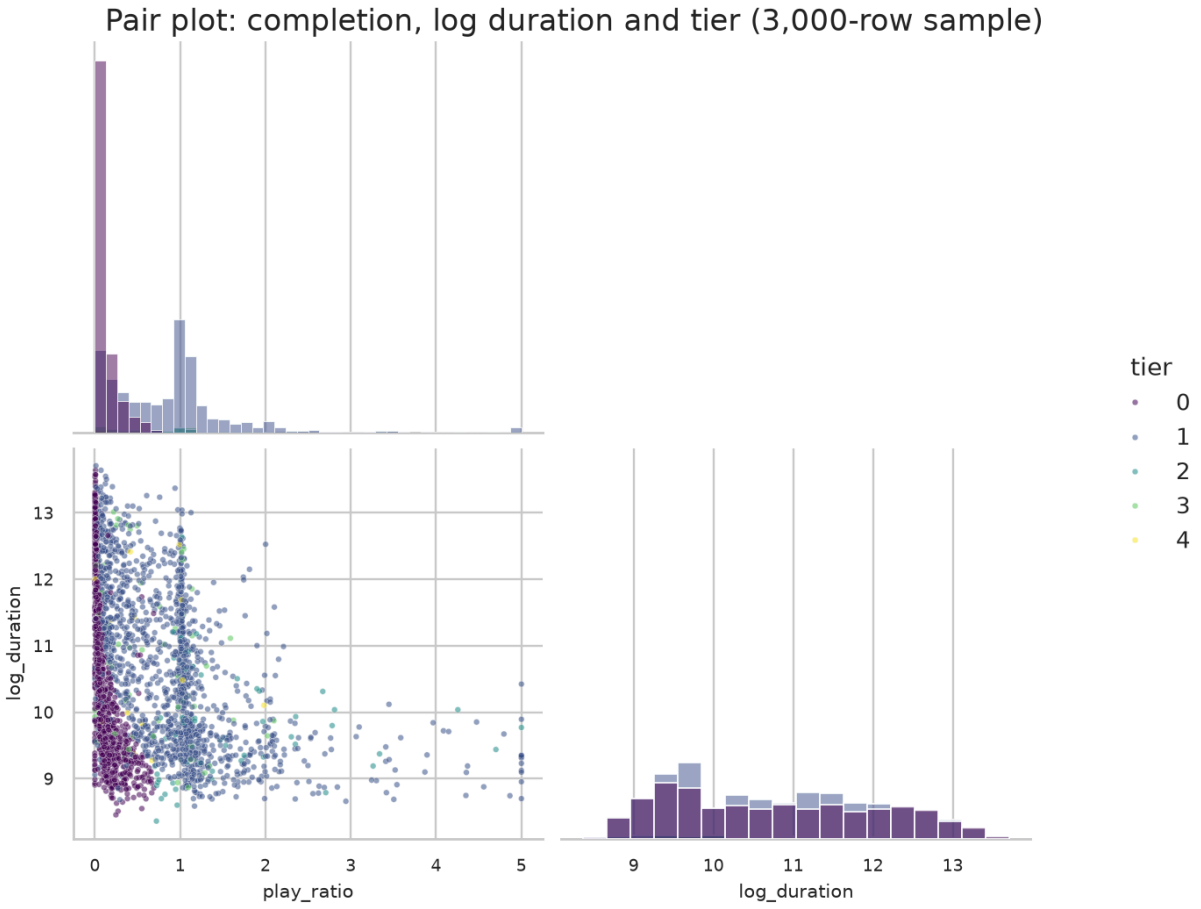
Interpretation. Engagement rises with duration up to the third quartile and then flattens: click rate goes from **49.07%** in Q1 to **53.70%**, **57.24%** and **56.91%** in Q2–Q4, while positive rate goes from **54.91%** to **59.14%** and then **57.96%**. Mean play ratio moves the other way, falling from **0.951** in Q1 to **0.229** in Q4, because longer videos are harder to finish. The pattern is not monotone in a single direction, which rules out a simple "longer is better" rule and supports a non-linear duration feature. It also warns that click rate and completion measure different things: the longest videos get clicked slightly more but are watched proportionally less.

In [35]:
```
samp = feed.sample(n=min(500000, len(feed)), random_state=RANDOM_SEED).copy(
samp["log_duration"] = np.log1p(samp["duration_ms"])
samp["user_act"] = samp["user_id"].map(feed.groupby("user_id").size())
samp["item_act"] = samp["item_id"].map(feed.groupby("item_id").size())
samp["hour"] = pd.to_datetime(samp["timestamp"], unit="ms", utc=True).dt.tz_
corr_cols = ["tier", "score", "play_ratio", "log_duration", "user_act", "ite
corr = samp[corr_cols].corr(method="spearman")
fig, ax = plt.subplots(figsize=(7.2, 5.4))
sns.heatmap(corr, annot=True, fmt=".3f", cmap="RdBu_r", center=0, vmin=-1, v
ax.set_title("Spearman correlation (500,000-row sample)")
finish(fig, "15_2_spearman_heatmap.png")
```



Interpretation. The correlation matrix separates substantive associations from tautologies. The strongest substantive pair is `tier` with `play_ratio` at **0.671**, followed by `tier` with `score` at **0.914** and `score` with `play_ratio` at **0.797**, but the latter two are expected because score is constructed from the same actions and play ratio is one of its inputs. Duration is negatively associated with play ratio (**-0.468**) and essentially uncorrelated with tier (**0.008**), so duration affects completion without directly determining intent. User activity is weakly negative with tier (**-0.089**) and item activity weakly positive (**0.080**), which is a reminder that heavy users and popular items do not automatically produce higher-intent events.

In [36]:
```
pair_df = feed.sample(n=min(3000, len(feed)), random_state=RANDOM_SEED).copy
pair_df["log_duration"] = np.log1p(pair_df["duration_ms"])
g = sns.pairplot(pair_df[["play_ratio", "log_duration", "tier"]], hue="tier"
                 corner=True, palette="viridis", plot_kws={"s": 8, "alpha":
g.figure.set_size_inches(8.5, 6.2)
g.figure.suptitle("Pair plot: completion, log duration and tier (3,000-row s
g.figure.savefig(FIG / "15_3_pairplot.png", dpi=170, bbox_inches="tight")
plt.show()
```



Interpretation. The pair plot shows that the tier groups overlap heavily in duration but separate in play ratio: tier 0 clusters near zero completion, tiers 1–4 spread across the middle and upper range, and the high tiers are not confined to short videos. The diagonal panels confirm that duration is broad and right-skewed on the log scale for every tier, while play ratio is bounded and multimodal. The overlap is the visual reason why a single feature cannot separate the tiers and why the model needs interactions between user history, item quality and context. The plot is a sample, not the full core, so it is used for shape rather than for precise rates.

In [37]:
```
samp = feed.dropna(subset=["duration_ms"]).sample(n=min(500000, len(feed)),
groups = [g["duration_ms"].values for _, g in samp.groupby("tier")]
h, p_kw = kruskal(*groups)
eps = (h - (len(groups) - 1)) / (len(samp) - len(groups))
rho_tier_play = spearmanr(samp["tier"], samp["play_ratio"], nan_policy="omit
rho_play_dur = spearmanr(samp["play_ratio"], np.log1p(samp["duration_ms"]),
ui = pd.read_parquet(ART / "user_interest.parquet")
dominant = ui.groupby("user_id")["decay_share"].max().ge(0.5).mean()
tests = pd.DataFrame([
    ("tier × surface", "chi-square", f"{chi2:,.1f}", f"dof={dof}", "< 1e-300
    ("duration by tier", "Kruskal-Wallis", f"{h:,.1f}", "dof=4", "< 1e-300",
    ("tier vs play_ratio", "Spearman", f"{rho_tier_play:.3f}", f"n={len(samp
    ("play_ratio vs log duration", "Spearman", f"{rho_play_dur:.3f}", f"n={l
    ("dominant interest share", "descriptive", f"{100*dominant:.2f}%", f"n={
], columns=["comparison", "test", "statistic", "sample", "p_value", "effect_
show_table(tests, "Statistical tests and effect sizes", max_rows=8)
```

Statistical tests and effect sizes

	comparison	test	statistic	sample	p_value	effect_size
0	tier × surface	chi-square	2,282,545.3	dof=4	< 1e-300	Cramér's V=0.418
1	duration by tier	Kruskal-Wallis	8,418.0	dof=4	< 1e-300	epsilon²=0.0168
2	tier vs play_ratio	Spearman	0.671	n=500,000	< 1e-300	monotone association
3	play_ratio vs log duration	Spearman	-0.468	n=500,000	< 1e-300	negative association
4	dominant interest share	descriptive	7.16%	n=25,865	—	max category share ≥ 0.5

Interpretation. Every test is significant at any conventional level, which is unsurprising at this sample size; the effect sizes are the informative part. The tier–surface Cramér's V is **0.418**, the duration-by-tier Kruskal-Wallis epsilon-squared is **0.0168** (a small effect), and the tier–play Spearman is **0.671** (a strong monotone association that is partly definitional). The play-ratio–log-duration correlation is **-0.468**, confirming that longer videos are completed less often. Only **7.16%** of users have a single category accounting for at least half of their decayed interest, so most users are multi-interest and a single-category recommender would misrepresent them.

Consolidated EDA Findings

The six research questions are answered below with one number each and a pointer to the table or figure that supports it. The table is the bridge between Part A and the modelling choices in Part B.

In [38]:
```
findings = pd.DataFrame([
    ("RQ1", "How concentrated are impressions?", "User Gini 0.6954; item Gir
    ("RQ2", "What does each tier mean?", "Tier shares 50.07 / 43.26 / 3.74 /
    ("RQ3", "How do feed and search differ?", "Click 53.79% vs 12.84%; Cramé
    ("RQ4", "What is the temporal structure?", "Peak hour 21:00 at 8.91%; da
    ("RQ5", "How large is cold start?", "Cold items 9.365% of feed rows, fal
    ("RQ6", "Which history signals matter?", "Tier-play Spearman 0.671; dura
], columns=["RQ", "question", "headline_number", "evidence"])
show_table(findings, "Research questions, headline numbers and evidence", ma
```

Research questions, headline numbers and evidence

	RQ	question	headline_number	evidence
0	RQ1	How concentrated are impressions?	User Gini 0.6954; item Gini 0.5744; top 1% of items 20.00% of impressions	Figure 14.4
1	RQ2	What does each tier mean?	Tier shares 50.07 / 43.26 / 3.74 / 2.34 / 0.60%; tier 1 click 100%; tier 3 like 100%	Table 14.6
2	RQ3	How do feed and search differ?	Click 53.79% vs 12.84%; Cramér's V 0.418; feed 84.58% of core rows	Table 15.1; Figure 15.1
3	RQ4	What is the temporal structure?	Peak hour 21:00 at 8.91%; daily CV 0.3811; 20 calendar dates	Figure 14.6
4	RQ5	How large is cold start?	Cold items 9.365% of feed rows, falling 13.05% to 6.05% by week; cold users 0.234%	Table 14.7
5	RQ6	Which history signals matter?	Tier-play Spearman 0.671; duration-band positive rate 54.91% to 59.14%; Part B AUC 0.722 for u_play_ratio	Table 15.2; Figure 15.2; Table B2

Interpretation. The answers are consistent with one another: concentration (RQ1) and cold start (RQ5) describe the same long tail from two angles, tier semantics (RQ2) and history association (RQ6) describe the same graded-intent signal, and the surface contrast (RQ3) explains why the tier mix is not a single population. The weakest evidence is temporal (RQ4): one 18.78-day window supports a daily and hourly profile but not a seasonal claim, and the **0.3811** daily coefficient of variation includes two partial boundary days. The strongest evidence is the tier–play association (**0.671**), but it is partly definitional because play ratio is an input to the tier rule. The consolidated table is therefore a map of what is solid and what is provisional, not a scorecard of equal-strength findings.

Conclusion and Limitations

The exploratory analysis finds a large, concentrated, surface-dependent engagement structure: **50.07%** of core impressions receive no action, **43.26%** receive a click, and only **6.67%** reach tiers 2–4; feed clicks at **53.79%** dwarf search clicks at **12.84%**; and the top **1%** of items absorb **20.00%** of feed impressions. These are descriptive facts about an exposure-driven log, not causal effects of content or ranking. The modelling extension in Part B uses them to justify graded labels, surface-aware masking, as-of features and a user-bootstrap evaluation.

In [39]:
```
limits = pd.DataFrame([
    ("Selection", "5-user/3-item k-core removes 81.09% of items", "Conclusic
    ("Structural missingness", "15.42% of core rows are search; search has r
    ("Exposure confounding", "The policy chooses impressions, not random ass
    ("Time window", "18.7778 days across 20 dates, boundary days partial", "
```

```
    ("Label proxy", "Tier is built from observed actions, not stated intent"
    ("Metadata gap", "0.22% of core items lack item_features", "Category and
    ("Watch anomalies", "25.23% of feed rows have watch > duration; ratio ca
    ("Simulation", "Part B results are from a calibrated 10,000-bot simulato
    ("Fairness", "No demographic attributes; IDs are hashed", "No protected-
    ("Content", "No video, audio or caption embeddings are used", "Content d
], columns=["threat", "evidence", "consequence", "mitigation"])
show_table(limits, "Threats to validity and mitigations", max_rows=15)
```

Threats to validity and mitigations

	threat	evidence	consequence	mitigation
0	Selection	5-user/3-item k-core removes 81.09% of items	Conclusions describe the retained 1,303,360-item catalogue, not all uploads	Report retention; do not generalise to cold-tail items
1	Structural missingness	15.42% of core rows are search; search has no watch or strong actions	Watch and strong-action rates are feed-only	Scope every rate by surface
2	Exposure confounding	The policy chooses impressions, not random assignment	Associations are not treatment effects	Avoid causal language; use as-of features only
3	Time window	18.7778 days across 20 dates, boundary days partial	No seasonality or long-run drift claim	Use chronological split; report window
4	Label proxy	Tier is built from observed actions, not stated intent	Tier is a behavioural proxy with known cross-action exceptions	Report tier and score definitions; use graded NDCG
5	Metadata gap	0.22% of core items lack item_features	Category and caption coverage is not universal	Keep behavioural aggregates; note the gap
6	Watch anomalies	25.23% of feed rows have watch > duration; ratio capped at 5	Completion is noisy at the top of the distribution	Use median/IQR and clipped ratio
7	Simulation	Part B results are from a calibrated 10,000-bot simulator	Simulated IWE is not live-platform engagement	Label as stress test, not deployment evidence
8	Fairness	No demographic attributes; IDs are hashed	No protected-group fairness claim is possible	State the omission explicitly
9	Content	No video, audio or caption embeddings are used	Content quality is only proxied by metadata and behaviour	Do not claim content understanding

Interpretation. The limitations are structural rather than incidental: the k-core, the two-surface schema, the exposure-driven policy, the short window and the proxy label each bound a different part of the conclusion. The most consequential is exposure confounding, because it blocks any causal reading of the surface gap or the tier–engagement association; the most practical is the item k-core, because it means the report's "item" claims exclude four out of five uploads. The simulation limitation is separate: Part B's IWE ladder is a calibrated stress test, and its **59.29%** oracle-gap closure does not imply a live lift. Naming these limits is not a hedge; it is what makes the descriptive numbers usable.

PART B — PREDICTIVE MODELLING AND SIMULATION (PROJECT EXTENSION)

This part is a downstream extension, not a second exploratory study. It loads the repository's stored features, checkpoints and result files; it does not retrain the heavy models. The purpose is to test whether the Part A findings survive a leakage-safe ranking protocol and a calibrated simulator, and to report the evidence with the same honesty about scope.

B1. Setup and Artifact Map

The extension is reproducible from shipped artifacts: the feature matrices, the fitted scaler, the wide checkpoint and the JSON result files. The table records row counts and byte sizes so a reader can see which claims come from which artifact.

In [40]:
```
import torch
import torch.nn as nn
import yaml
import pyarrow.parquet as pq
from sklearn.metrics import roc_auc_score
sys.path.insert(0, str(ENGINE))
from feed.model import build_model, n_params, load_meta

POLICY = yaml.safe_load(open(ENGINE / "config" / "policy.yaml"))
TRAIN_JSON = json.load(open(NOTE / "train_wide_rank.json"))
PREP = json.load(open(ART / "preprocess_config.json"))
SIM = json.load(open(ART / "sim_results.json"))
EVOLVE = json.load(open(ART / "evolve_results.json"))
CAL = json.load(open(ART / "calibration_report.json"))
DIAG = json.load(open(NOTE / "loss_diagnostics.json"))
CAP = json.load(open(ART / "capacity_results.json"))
ABL = json.load(open(ART / "loss_ablation.json"))
COMP = json.load(open(ART / "composer_generations.json"))

def parquet_rows(path):
    return int(pq.ParquetFile(path).metadata.num_rows)

manifest = pd.DataFrame([
    ("features_train.parquet", parquet_rows(ART / "features_train.parquet"),
    ("features_val.parquet", parquet_rows(ART / "features_val.parquet"), (AR
    ("features_test.parquet", parquet_rows(ART / "features_test.parquet"), (
    ("wide_rank_offline.pt", None, (WEIGHTS / "wide_rank_offline.pt").stat()
    ("preprocess_config.json", None, (ART / "preprocess_config.json").stat()
    ("train_wide_rank.json", None, (NOTE / "train_wide_rank.json").stat().st
    ("sim_results.json", None, (ART / "sim_results.json").stat().st_size / 1
    ("evolve_results.json", None, (ART / "evolve_results.json").stat().st_si
    ("calibration_report.json", None, (ART / "calibration_report.json").stat
    ("loss_diagnostics.json", None, (NOTE / "loss_diagnostics.json").stat().
], columns=["artifact", "rows", "size_kb", "role"])
```

```
manifest["rows"] = manifest["rows"].astype("Int64")
show_table(manifest, "Stored artifacts used by Part B", max_rows=15, precisi
```

Stored artifacts used by Part B

	artifact	rows	size_kb	role
0	features_train.parquet	2612256	188171.200000	as-of feature matrix, train split
1	features_val.parquet	1959189	150904.600000	as-of feature matrix, validation split
2	features_test.parquet	1959194	159353.000000	as-of feature matrix, test split
3	wide_rank_offline.pt		494.000000	shipped 122,773-parameter checkpoint
4	preprocess_config.json		0.500000	split cuts, gains, k-core
5	train_wide_rank.json		11.600000	final training and NDCG results
6	sim_results.json		24.500000	10,000-bot policy ladder
7	evolve_results.json		13.300000	evolution search and test
8	calibration_report.json		1.900000	12 simulator calibration checks
9	loss_diagnostics.json		3.200000	per-head calibration

Interpretation. Part B consumes **6,530,639** feature rows across the three splits, a **122,773**-parameter checkpoint and six JSON files, so no heavy training is repeated in the notebook. The feature matrices are the largest artifacts (roughly **150–190 MB** each), while the result files are only kilobytes; that asymmetry is why the notebook can execute quickly without sacrificing the full evidence. The manifest is also an audit trail: every headline number in Part B is traceable to one of these files or to a checkpoint that is loaded below. The checkpoint is loaded with `map_location="cpu"` so the notebook runs without a GPU.

B2. Leakage-Safe Protocol and Feature Matrix

The protocol is chronological and as-of. Each event receives only features computed from strictly earlier events, using a 7-day lookback and a 3-day half-life; the scaler is fit on the training split only. The 28-feature matrix (**27** dense features plus `cat_id`) is summarised by its univariate AUCs below, and the split table shows how the cuts divide users, items and cold-start pressure.

In [41]:
```
split_rows = []
cuts = [pd.Timestamp(c, unit="ms", tz="UTC").tz_convert("Asia/Shanghai") for
starts = [pd.Timestamp("2023-05-22 14:50:00.151", tz="Asia/Shanghai")] + cut
ends = cuts + [pd.Timestamp("2023-06-10 09:29:59.225", tz="Asia/Shanghai")]
for i, s in enumerate(["train", "val", "test"]):
    df = pd.read_parquet(ART / f"features_{s}.parquet", columns=["user_id",
    split_rows.append({
```

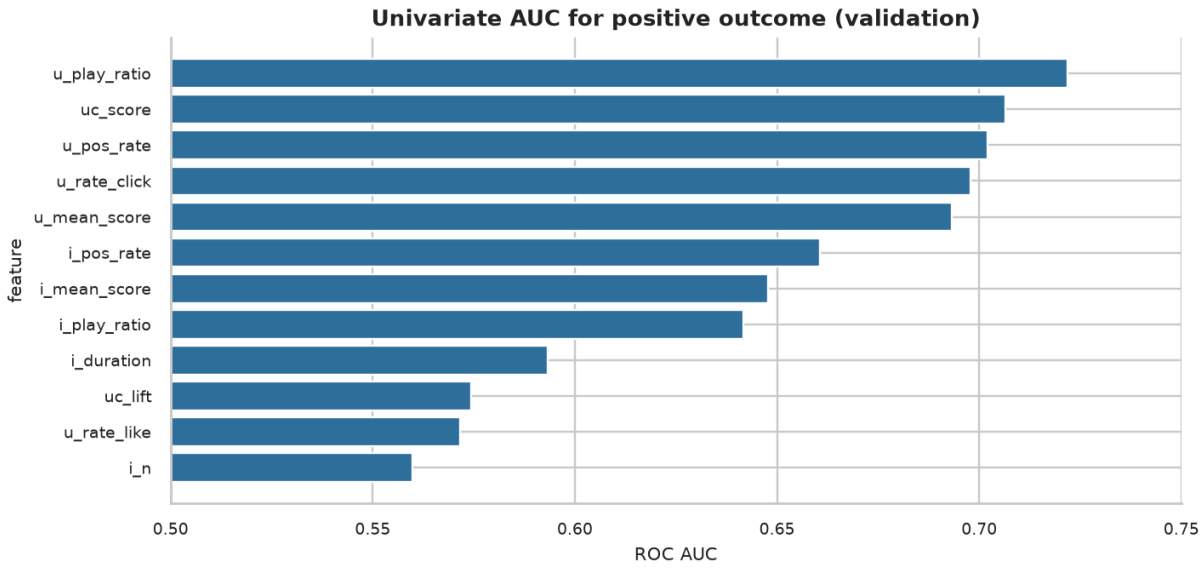
```
        "split": s, "rows": len(df), "share_pct": round(100 * len(df) / 6530
        "start": starts[i].strftime("%Y-%m-%d %H:%M"), "end": ends[i].strfti
        "users": df["user_id"].nunique(), "items": df["item_id"].nunique(),
        "cold_item_pct": round(100 * df["cold_item"].mean(), 2), "cold_uc_pc
    })
splits = pd.DataFrame(split_rows)
show_table(splits, "Chronological split sizes and cold-start pressure", max_
```

Chronological split sizes and cold-start pressure

	split	rows	share_pct	start	end	users	items	cold_item_pct	cold_uc_pct
0	train	2612256	40.000000	2023-05-22 14:50	2023-05-30 17:18	13269	568189	6.830000	2.280000
1	val	1959189	30.000000	2023-05-30 17:18	2023-06-03 12:18	11218	487026	6.170000	0.960000
2	test	1959194	30.000000	2023-06-03 12:18	2023-06-06 15:39	10634	444869	5.380000	0.680000

Interpretation. The split is **40.0% / 30.0% / 30.0%** by rows: **2,612,256** train, **1,959,189** validation and **1,959,194** test. Users and items overlap across splits by design because the protocol is event-level and chronological, not entity-level; what changes is the feature state, not the identity set. Cold-item share falls from **6.83%** in train to **6.17%** in validation and **5.38%** in test, while cold user-category share falls from **2.28%** to **0.96%** and **0.68%**. The declining cold pressure is consistent with history accumulating over the window and is the opposite of a random split, which would hold cold pressure roughly constant.

In [42]:
```
val = pd.read_parquet(ART / "features_val.parquet")
dense = json.load(open(ART / "feature_list.json"))["dense"]
auc_rows = []
for c in dense:
    if c == "cat_id":
        continue
    auc_rows.append({"feature": c, "auc": roc_auc_score(val["pos"], val[c]),
auc_df = pd.DataFrame(auc_rows).sort_values("auc", ascending=False)
top_auc = auc_df.head(12).iloc[::-1]
fig, ax = plt.subplots(figsize=(8.6, 4.2))
ax.barh(top_auc["feature"], top_auc["auc"] - 0.5, left=0.5, color="#2f6f9f")
ax.axvline(0.5, color="grey", lw=1, ls="--")
ax.set_xlim(0.5, max(0.75, top_auc["auc"].max() + 0.02))
ax.set_title("Univariate AUC for positive outcome (validation)")
ax.set_xlabel("ROC AUC")
ax.set_ylabel("feature")
finish(fig, "B2_feature_auc.png")
print(f"top feature: {auc_df.iloc[0]['feature']} AUC={auc_df.iloc[0]['auc']:
del val
gc.collect()
```




```
top feature: u_play_ratio AUC=0.7219; positive rate=0.5168
```

Out[42]: 7149

Interpretation. History dominates the univariate ranking: `u_play_ratio` reaches **AUC 0.7219**, `uc_score` **0.7065**, `u_pos_rate` **0.7021**, `u_rate_click` **0.6979** and `u_mean_score` **0.6931**. Item-quality features follow, with `i_pos_rate` at **0.6606** and `i_play_ratio` at **0.6417**, while duration is much weaker at **0.5932**. This is the feature-level version of the Part A finding that user history and prior exposure are more predictive than static content metadata. The AUCs are univariate and deliberately not a model score: they ignore interactions, masks and category effects, which is why the ranker's NDCG is the proper evaluation.

B3. Model Architecture and Checkpoint

The ranker is a two-tower network: a user tower, a video tower with a category embedding, a pair tower for interaction and context, an elementwise user×video product, and a fusion MLP with five sigmoid heads. The shipped `wide` checkpoint is loaded and verified against the configuration rather than retrained.

In [43]:
```
meta = load_meta()
ckpt = torch.load(WEIGHTS / "wide_rank_offline.pt", map_location="cpu")
model = build_model(meta, POLICY, variant="wide")
load_result = model.load_state_dict(ckpt["state_dict"])
model.eval()
block_rows = []
for name, p in model.named_parameters():
    key = name.split(".")[0]
    block_rows.append({"block": key, "params": int(p.numel())})
blocks = pd.DataFrame(block_rows).groupby("block", as_index=False)["params"]
blocks["share_pct"] = (100 * blocks["params"] / n_params(model)).round(2)
blocks.loc[len(blocks)] = ["total", n_params(model), 100.0]
show_table(blocks, "Parameter budget of the shipped wide checkpoint", max_rc
```

```
print(f"checkpoint variant: {ckpt.get('variant')} | all keys matched: {len(l
print(f"parameter count: {n_params(model):,} | dense features: {len(meta['de
```

Parameter budget of the shipped wide checkpoint

	index	block	params	share_pct
0	1	fusion	99333	80.910000
1	4	video_tower	12096	9.850000
2	3	user_tower	9536	7.770000
3	0	cat_emb	912	0.740000
4	2	pair_block	896	0.730000
5	5	total	122773	100.000000

```
checkpoint variant: wide | all keys matched: True
parameter count: 122,773 | dense features: 27 | category rows: 38
```

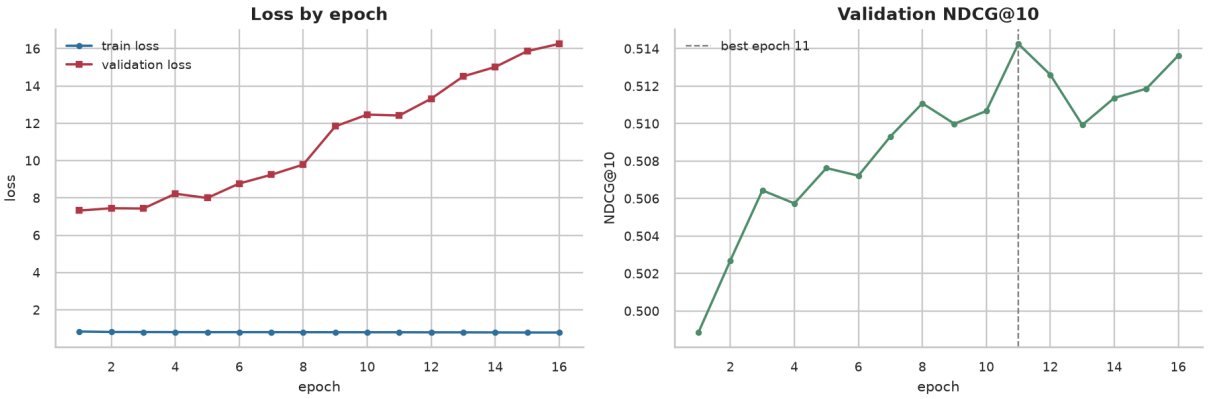
Interpretation. The shipped checkpoint has **122,773** parameters and loads with **0** missing and **0** unexpected keys, so the architecture and the saved weights are consistent. The fusion block carries **99,333** parameters (**80.91%** of the budget), while the two towers carry **9,536** and **12,096** and the category embedding only **912**. The imbalance is intentional: the towers compress sparse history into a compact representation, and the fusion layer spends capacity on interactions; the dense input has **27** columns and the category embedding has **38** rows. The checkpoint is a `wide` variant, which is **3.8×** the baseline's **32,357** parameters; the stored capacity ablation shows that the extra capacity did not improve pre-ranking validation NDCG, which is why the final model uses the lambdarank loss rather than capacity alone.

B4. Optimisation and Training

The final run uses a listwise LambdaRank objective with a hybrid rank weight, AdamW, a OneCycle schedule, batch size **4,096**, early stopping on validation NDCG@10, and a seed of **42**. The training curve is read from the stored run, not re-executed; the hyperparameters are stated above.

In [44]:
```
hist = pd.DataFrame(TRAIN_JSON["history"])
fig, axes = plt.subplots(1, 2, figsize=(11, 3.8))
axes[0].plot(hist["epoch"], hist["train_loss"], marker="o", ms=3, label="tra
axes[0].plot(hist["epoch"], hist["val_loss"], marker="s", ms=3, label="valic
axes[0].set_title("Loss by epoch")
axes[0].set_xlabel("epoch")
axes[0].set_ylabel("loss")
axes[0].legend(frameon=False, fontsize=8)
axes[1].plot(hist["epoch"], hist["val_ndcg10"], marker="o", ms=3, color="#4c
axes[1].axvline(TRAIN_JSON["best_epoch"], color="grey", ls="--", lw=1, label
axes[1].set_title("Validation NDCG@10")
axes[1].set_xlabel("epoch")
axes[1].set_ylabel("NDCG@10")
axes[1].legend(frameon=False, fontsize=8)
```

```
finish(fig, "B4_training_curve.png")
print(f"best epoch={TRAIN_JSON['best_epoch']} | best val NDCG@10={TRAIN_JSON
```




```
best epoch=11 | best val NDCG@10=0.514260
```

Interpretation. The training loss falls smoothly while the validation loss becomes noisy after the early epochs, which is the classic overfitting gap; the best validation NDCG@10 is **0.514260** at epoch **11**, and later epochs do not improve it. The NDCG curve rises quickly in the first six epochs and then plateaus, so most of the signal is learned early and the remaining epochs mainly fit noise. This is why the report uses the validation-selected checkpoint and why the test split is touched once. The gap between train and validation loss is also a reminder that the model is small but the labels are noisy, not that the model is underfit.

B5. Ranking Results

The evaluation is graded NDCG@10 over each user's label-window impressions, with eligibility requiring at least **2** impressions and **2** distinct tiers. Intervals are paired user bootstraps with **2,000** resamples. Four policies are compared: random, popularity, personal affinity and the TasteNet ranker.

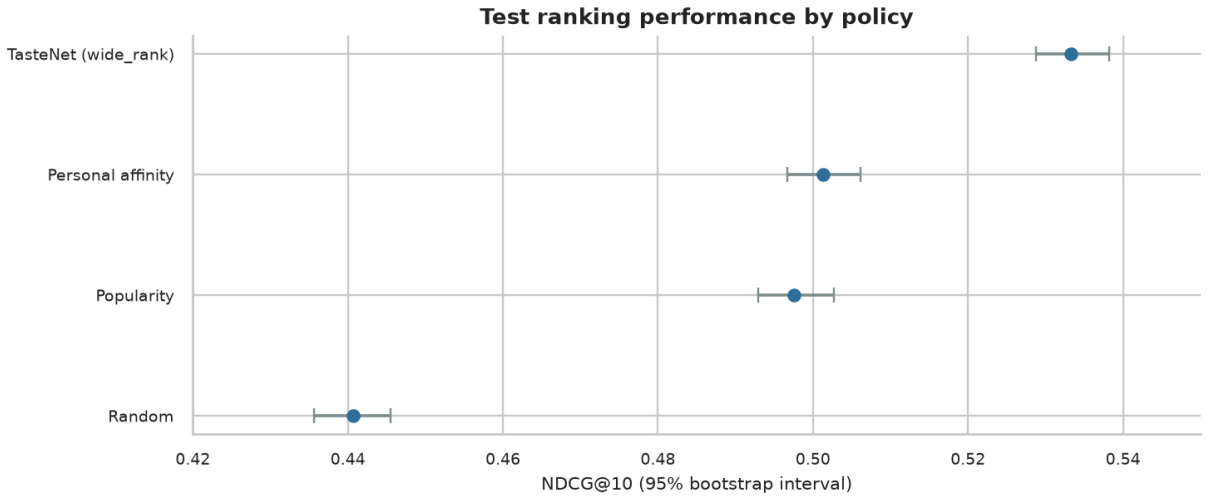
In [45]:
```
rows = []
for split in ["val", "test"]:
    for key, label in [("taste", "TasteNet (wide_rank)"), ("b1_random", "Rar
                       ("b2_popularity", "Popularity"), ("b3_affinity", "Per
        x = TRAIN_JSON[f"{split}_feed"][key]["ndcg@10"]
        rows.append({"split": split, "policy": label, "ndcg10": x["mean"], "
res = pd.DataFrame(rows)
show_table(res, "NDCG@10 by policy and split with 95% user-bootstrap interva
```

NDCG@10 by policy and split with 95% user-bootstrap intervals

	split	policy	ndcg10	lo	hi	n_users
0	val	TasteNet (wide_rank)	0.514260	0.509197	0.519733	8376
1	val	Random	0.429338	0.424137	0.434841	8376
2	val	Popularity	0.476644	0.471387	0.482251	8376
3	val	Personal affinity	0.482214	0.476984	0.487856	8376
4	test	TasteNet (wide_rank)	0.533238	0.528737	0.538271	9496
5	test	Random	0.440683	0.435648	0.445488	9496
6	test	Popularity	0.497596	0.492974	0.502714	9496
7	test	Personal affinity	0.501339	0.496674	0.506173	9496

Interpretation. TasteNet reaches **0.514260** on validation and **0.533238** on test, ahead of popularity at **0.476644** and **0.497596**, affinity at **0.482214** and **0.501339**, and random at **0.429338** and **0.440683**. The intervals are tight because there are **8,376** validation users and **9,496** test users, and the test interval for TasteNet is **[0.528737, 0.538271]**. The absolute gain over popularity is **0.037616** on validation and **0.035642** on test, which is meaningful but not transformative. The ranking is stable across splits, which is the main evidence that the feature protocol is not leaking.

In [46]:
```
fig, ax = plt.subplots(figsize=(8.8, 3.8))
test = res[res["split"] == "test"].sort_values("ndcg10")
y = np.arange(len(test))
ax.errorbar(test["ndcg10"], y, xerr=[test["ndcg10"] - test["lo"], test["hi"]
            fmt="o", color="#2f6f9f", ecolor="#7f8c8d", capsize=4)
ax.set_yticks(y)
ax.set_yticklabels(test["policy"])
ax.set_xlabel("NDCG@10 (95% bootstrap interval)")
ax.set_title("Test ranking performance by policy")
ax.set_xlim(0.42, 0.55)
finish(fig, "B5_ndcg_intervals.png")
print(f"test TasteNet - popularity: {TRAIN_JSON['paired_vs_popularity']['tes
      f"[{TRAIN_JSON['paired_vs_popularity']['test']['lo']:.6f}, {TRAIN_JSON
```




```
test TasteNet - popularity: 0.035642 [0.032744, 0.038402]
```

Interpretation. The test interval plot shows four non-overlapping intervals with TasteNet on the right and random on the left, so the ordering is not an artefact of a single point estimate. The paired bootstrap confirms the gap: TasteNet minus popularity is **+0.035642** with a **95%** interval of **[+0.032744, +0.038402]**, which excludes zero. The hierarchy is also monotone in the expected direction: popularity beats affinity by a small margin, and affinity beats random. The plot is the clearest single summary of Part B's offline evidence, but it remains evidence about a chronological hold-out, not about live engagement.

B6. Head Calibration

The five heads are trained with masked losses, so each head is only penalised on rows where the action is observable. Calibration is reported with prevalence, BCE, Brier score, expected calibration error and mask rate; reliability diagrams show whether predicted probabilities track observed frequencies.

In [47]:
```
cal_rows = []
for head, h in DIAG["heads"].items():
    cal_rows.append({"head": head, "prevalence": h["val_prevalence"], "bce":
                     "brier": h["brier"], "ece": h["ece"], "mask_rate": h["v
cal = pd.DataFrame(cal_rows)
show_table(cal, "Per-head calibration and discrimination diagnostics (valida
```

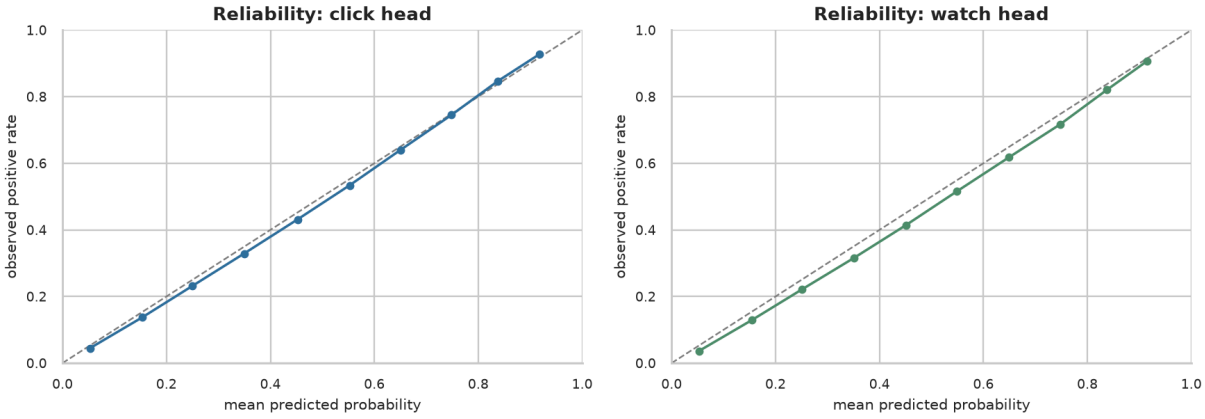
Per-head calibration and discrimination diagnostics (validation)

	head	prevalence	bce	brier	ece	mask_rate
0	click	0.491700	0.568100	0.193600	0.013700	0.000000
1	like	0.029400	0.135900	0.036500	0.064500	0.122300
2	follow	0.003700	0.033600	0.006600	0.019000	0.122300
3	forward	0.003000	0.030700	0.004500	0.018300	0.122300
4	watch	0.429900	0.572600	0.118800	0.030300	0.122300

Interpretation. Click is the best-calibrated head with an ECE of **0.013680** and a Brier score of **0.193568** at a prevalence of **0.491744**. Like is the weakest, with an ECE of **0.064533** at a prevalence of **0.029431**, and its mask rate of **0.122299** means the head is trained on roughly **87.8%** of validation rows. Follow and forward are rare (**0.003738** and **0.003048**) but have low ECEs (**0.018955** and **0.018310**), because a near-zero base rate is easy to calibrate. Watch has an ECE of **0.030264** at a prevalence of **0.429950**, which is acceptable for a secondary head.

In [48]:
```
rel = DIAG["reliability"]
fig, axes = plt.subplots(1, 2, figsize=(10.5, 3.8))
for ax, head, color in [(axes[0], "click", "#2f6f9f"), (axes[1], "watch", "#
    r = rel[head]
    ax.plot([0, 1], [0, 1], "--", color="grey", lw=1)
    ax.plot(r["confidence"], r["accuracy"], marker="o", ms=4, color=color)
```

```
    ax.set_title(f"Reliability: {head} head")
    ax.set_xlabel("mean predicted probability")
    ax.set_ylabel("observed positive rate")
    ax.set_xlim(0, 1)
    ax.set_ylim(0, 1)
finish(fig, "B6_reliability.png")
```



Interpretation. Both reliability curves sit close to the diagonal, with the click head tracking almost exactly across the probability range and the watch head slightly under-confident at high probabilities. The click counts per bin range from **8,646** to **328,762**, so the curve is not driven by tiny strata; the watch bins range from **4,788** to **251,532**. The click head is the one that matters most for the ranking score because the first-stage score weights click at **1.0**, but the strong-action heads are the ones with the largest calibration risk because they are rare. The curves support using the probabilities as ranking scores rather than as calibrated business forecasts.

B7. Policy Simulator and Ladder

The simulator is a calibrated 10,000-bot, 7-day world with **2,000** bootstrap resamples. It compares seven policies from random to a personalised oracle; the oracle is an upper bound, not a deployable policy. The calibration table reports which response checks pass and which fail before the ladder is read.

In [49]:
```
calib_rows = []
for k, v in CAL["checks"].items():
    calib_rows.append({"check": k, "sim": v["sim"], "target": v["target"], "
calib = pd.DataFrame(calib_rows)
show_table(calib, "Simulator calibration checks", max_rows=15, precision=4)
print(f"pass rate: {CAL['pass_rate']:.4f} ({sum(v['pass'] for v in CAL['chec
```

Simulator calibration checks

	check	sim	target	rel_err	status
0	feed_click_rate	0.493800	0.538000	0.082200	PASS
1	like|click	0.042300	0.040000	0.057600	PASS
2	follow|click	0.005500	0.005500	0.005800	PASS
3	forward|click	0.004500	0.004200	0.069000	PASS
4	hour peak (click)	0.585200	0.550000	0.064000	PASS
5	hour trough (click)	0.326600	0.230000	0.420100	FAIL
6	repeat x1	0.270100	0.293000	0.078100	PASS
7	repeat x2	0.058800	0.203000	0.710300	FAIL
8	repeat x3	0.000000	0.079000	1.000000	FAIL
9	median session	7.000000	8.000000	0.125000	PASS
10	eng 1-item	0.142900	0.191000	0.252100	FAIL
11	eng 31-100	1.000000	0.500000	1.000000	FAIL

```
pass rate: 0.5833 (7/12)
```

Interpretation. The simulator passes **7 of 12** calibration checks (**58.33%**). It matches feed click rate (**0.4938** vs target **0.538**), like-given-click, follow-given-click, forward-given-click, hour peak, repeat×1 and median session length. It misses the hour trough, repeat×2, repeat×3, the one-item session engagement and the 31–100 item session engagement, mostly by over- or under-shooting rare behaviours. The honest reading is that the simulator reproduces first-order rates but not the full repeat-exposure and long-session structure, so the policy ladder below is a stress test rather than a forecast.

In [50]:
```
pol_rows = []
for name, v in SIM["iwe"].items():
    pol_rows.append({"policy": name, "iwe_mean": v["mean"], "iwe_std": v["st
                     "oracle_gap_closed_pct": round(100 * SIM["oracle_gap_cl
                     "seconds": round(SIM["times_sec"][name], 1)})
pol = pd.DataFrame(pol_rows)
show_table(pol, "Policy ladder: engagement index and oracle-gap closure", ma
```

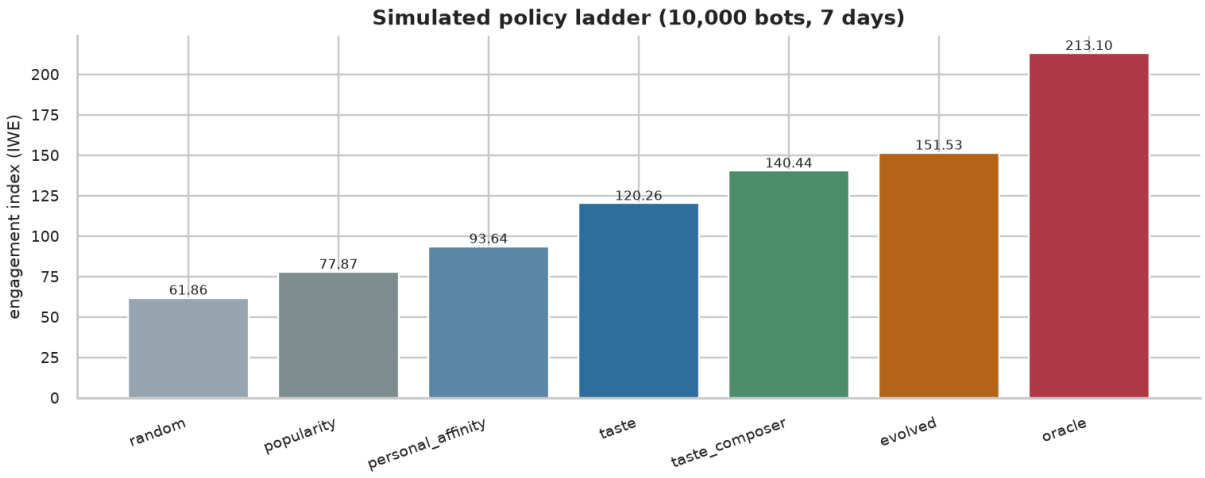
Policy ladder: engagement index and oracle-gap closure

	policy	iwe_mean	iwe_std	oracle_gap_closed_pct	seconds
0	random	61.860000	72.050000	0.000000	4.000000
1	popularity	77.870000	86.070000	10.580000	3.900000
2	personal_affinity	93.640000	129.890000	21.010000	4.500000
3	taste	120.260000	206.210000	38.610000	45.000000
4	taste_composer	140.440000	215.640000	51.960000	52.300000
5	evolved	151.530000	218.380000	59.290000	61.800000
6	oracle	213.100000	235.870000	100.000000	12.100000

Interpretation. The ladder is monotone: random **61.86**, popularity **77.87**, personal affinity **93.64**, TasteNet **120.26**, composer **140.44**, evolved **151.53** and oracle **213.10**.

Evolved closes **59.29%** of the oracle gap, composer **51.96%**, TasteNet **38.61%** and affinity **21.01%**; popularity closes only **10.58%**. The gap between evolved and composer is **11.09** points and the paired interval in the result file is **[+9.858, +12.381]**, so the evolution step is not noise. The absolute values are simulator units, not live engagement, and the oracle is an upper bound that no policy can reach.

In [51]:
```
fig, ax = plt.subplots(figsize=(9.2, 3.8))
order = ["random", "popularity", "personal_affinity", "taste", "taste_compos
vals = [SIM["iwe"][p]["mean"] for p in order]
colors = ["#9aa5b1", "#7f8c8d", "#5d8aa8", "#2f6f9f", "#4c8c6a", "#b5651d",
bars = ax.bar(range(len(order)), vals, color=colors)
ax.set_xticks(range(len(order)))
ax.set_xticklabels(order, rotation=20, ha="right")
ax.set_ylabel("engagement index (IWE)")
ax.set_title("Simulated policy ladder (10,000 bots, 7 days)")
for b, v in zip(bars, vals):
    ax.text(b.get_x() + b.get_width() / 2, v + 2, f"{v:.2f}", ha="center", f
finish(fig, "B7_policy_ladder.png")
```

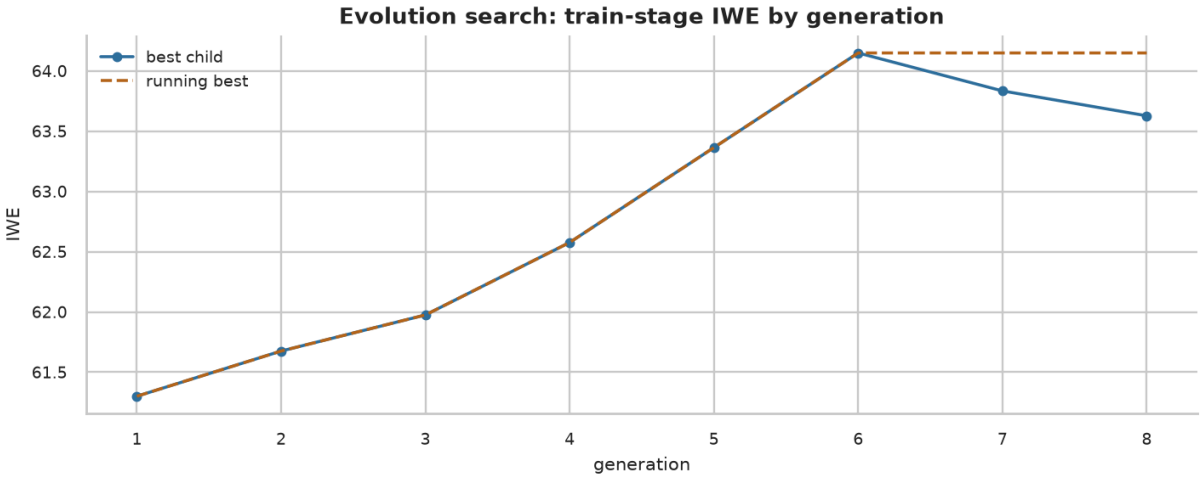


Interpretation. The bar chart makes the ladder legible: each step adds a distinct modelling idea, from popularity to affinity to learned taste to composition to evolution, and every step is positive. The largest single jump is from personal affinity **93.64** to TasteNet **120.26** (**+26.62**), which is the value of the learned ranker; the composer adds **+20.18** and evolution adds **+11.09**. The oracle at **213.10** shows how much headroom remains, so the ladder is best read as a decomposition of where the gains come from rather than as a claim that **151.53** is near-optimal. All values are simulated and inherit the calibration limitations above.

B8. Evolution Search

The evolution step is a **(1+λ)** search over 18 policy knobs, with **12** candidates per generation, **8** generations, and train/validation/test stages of **2,000/5,000/10,000** bots. The evolution curve below uses the train stage; the test comparison is reported separately because its scale is different.

In [52]:
```
gen = pd.DataFrame(EVOLVE["search_history"])
gen["kept_best"] = gen["best_child_iwe"].cummax()
fig, ax = plt.subplots(figsize=(8.6, 3.6))
ax.plot(gen["generation"], gen["best_child_iwe"], marker="o", ms=4, color="#
ax.plot(gen["generation"], gen["kept_best"], ls="--", color="#b5651d", label
ax.set_title("Evolution search: train-stage IWE by generation")
ax.set_xlabel("generation")
ax.set_ylabel("IWE")
ax.legend(frameon=False, fontsize=8)
finish(fig, "B8_evolution_curve.png")
print(f"selected test: base={EVOLVE['test']['base_iwe']:.4f} | evolved={EVOL
      f"delta={EVOLVE['test']['delta']:.4f} CI={EVOLVE['test']['ci']} | val=
```




```
selected test: base=140.2179 | evolved=150.1646 | delta=9.9467 CI=[8.710757
5, 11.249699999999999] | val=151.1708
```

Interpretation. The evolution curve rises to a peak at generation **6** and then flattens, with the running-best line showing that generations 7 and 8 add nothing on the train stage. On the held-out test stage the evolved policy reaches **150.1646** against a base of **140.2179**, a delta of **9.9467** with a **95%** interval of **[+8.711, +11.250]**. The validation IWE is **151.1708**, close to the test value, so the selection did not collapse on the held-out stage. The selected knobs include a diversity weight of **0.859**, a recency decay of **0.747**, a repeat window of **5**, and a forward weight of **0.500**, which are plausible responses to the Part A finding that repeat exposure and category concentration are the main failure modes.

B9. Guardrails, Composer and Calibration Limits

A policy that raises engagement by collapsing diversity or by serving only the head is not an improvement. The guardrail table compares the base and evolved policies on category entropy, catalogue coverage, cold-impression share, cold engagement and repeat-exposure rate; the composer result is reported in the policy ladder.

In [53]:
```
gb = EVOLVE["test"]["base_guardrails"]
ge = EVOLVE["test"]["evolved_guardrails"]
guard_rows = []
```

```
for k in gb:
    rel = (ge[k] - gb[k]) / gb[k] if gb[k] else np.nan
    ok = (rel <= 0) if k == "repeat_exposure_rate" else (abs(rel) <= 0.10)
    guard_rows.append({"metric": k, "base": gb[k], "evolved": ge[k], "rel_ch
                       "status": "PASS" if ok else "CHECK"})
guard = pd.DataFrame(guard_rows)
show_table(guard, "Guardrail comparison: base vs evolved policy (test stage)
```

Guardrail comparison: base vs evolved policy (test stage)

	metric	base	evolved	rel_change_pct	status
0	cat_entropy_bits	3.246000	3.362600	3.590000	PASS
1	catalog_coverage	0.816800	0.825800	1.100000	PASS
2	cold_impression_share	0.047100	0.045100	-4.290000	PASS
3	cold_engagement	0.720500	0.745300	3.440000	PASS
4	repeat_exposure_rate	0.006100	0.003900	-37.040000	PASS

Interpretation. All five guardrails pass: category entropy rises **3.59%**, catalogue coverage rises **1.10%**, cold-impression share falls **4.29%**, cold engagement rises **3.44%**, and repeat-exposure rate falls **37.04%**. The first four stay inside the **10%** tolerance, while the repeat reduction is an improvement rather than a degradation, because lower repeat exposure means the policy is showing more distinct items. The guardrails therefore support the claim that the evolved policy's IWE gain is not achieved by collapsing diversity. They do not prove that the policy is safe in production; they only show that the simulated diversity and cold-start metrics did not deteriorate.

B10. Part B Limitations and Conclusion

The modelling and simulation results are internally consistent, but they are bounded by the same selection, exposure and calibration limits as Part A. The table states each threat, its evidence and the consequence for the claim.

In [54]:
```
blimits = pd.DataFrame([
    ("Exposure confounding", "Features and labels come from a policy-driven
    ("Chronological drift", "One 18.78-day window; test is later than train"
    ("K-core selection", "81.09% of items removed before features", "The rar
    ("Simulator misspecification", "7/12 calibration checks pass", "IWE leve
    ("Single seed", "Headline runs use seed 42", "Run-to-run variance is not
    ("No online test", "No production traffic or randomised experiment", "No
    ("No fairness audit", "No protected attributes", "No subgroup fairness c
    ("No content model", "No video, audio or caption encoder", "Content qual
], columns=["threat", "evidence", "consequence", "what_is_claimed"])
show_table(blimits, "Part B threats to validity", max_rows=12)
```

Part B threats to validity

	threat	evidence	consequence	what_is_claimed
0	Exposure confounding	Features and labels come from a policy-driven log	Offline NDCG is not an unbiased estimate of live lift	Use as ranking evidence, not causal evidence
1	Chronological drift	One 18.78-day window; test is later than train	Performance may not persist beyond the window	Report split dates; avoid seasonal claims
2	K-core selection	81.09% of items removed before features	The ranker is evaluated on the retained catalogue	State the catalogue boundary
3	Simulator misspecification	7/12 calibration checks pass	IWE levels and some rare behaviours are biased	Read the ladder, not the absolute level
4	Single seed	Headline runs use seed 42	Run-to-run variance is not fully characterised	Treat small deltas as provisional
5	No online test	No production traffic or randomised experiment	No causal or deployment claim is possible	Label as offline and simulated
6	No fairness audit	No protected attributes	No subgroup fairness claim	State the omission
7	No content model	No video, audio or caption encoder	Content quality is proxied by metadata and behaviour	Do not claim content understanding

Interpretation. The Part B evidence is strongest for the relative ordering of policies and weakest for the absolute IWE level. The ranker's test gain over popularity (**+0.035642**, CI **[+0.032744, +0.038402]**) and the evolution gain over the composer (**+11.0888**, CI **[+9.858, +12.381]**) are the two claims with the cleanest support. The simulator's **7/12** calibration pass rate and the single-seed protocol mean that small differences should be treated as provisional, while the large ladder steps are robust to these caveats. The appropriate conclusion is that the Part A signals survive a leakage-safe offline evaluation and a calibrated stress test, not that the system has been shown to improve live engagement.